05 - MLE, MAP, Bayes

Pattern Recognition and Machine Learning, Lab 5

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23 October 2026

Recap

What do we know about the classes?

Bayesian task

- $p(x \mid k)$ **known**
- $p(k), W(k, d)$ **known**

We know **everything**.

Parzen windows

- $p(x \mid k)$ **unknown**
- **any** shape

Non-parametric: histogram, k-NN, kernels.

last lab

Today

- $p(x \mid k)$ **unknown**
- **known form** $p(x \mid \theta)$, unknown θ

Parametric: ML, MAP, Bayes.

today

! We estimate a few **parameters** instead of the whole density, e.g. $\theta = (\mu, \sigma^2)$ of a normal distribution.

The setting

Known form, unknown parameters:

$$p(x \mid k) = p(x \mid k, \theta_k)$$

Each class separately, as with Parzen: we drop k and write $p(x \mid \theta)$.

Training set, samples **i.i.d.**:

$$T = \{x_1, x_2, \ldots, x_N\}$$

Likelihood of the parameters:

$$L(\theta) = p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$$

Log-likelihood, same maximum:

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{N} \ln p(x_i \mid \theta)$$

> **!** $L(\theta)$ is a function of θ for **fixed data**, **not** a distribution over θ.

Likelihood: try it

μ 1.2

σ 1.6

☑ density ☑ log-likelihood

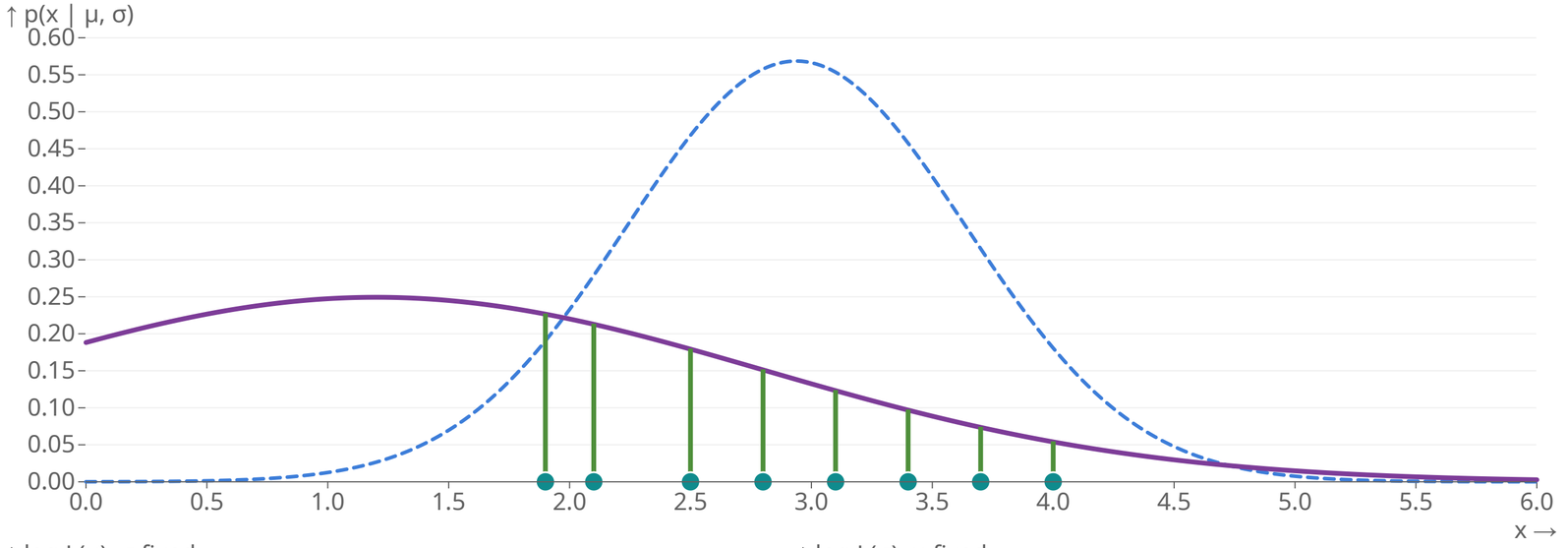
☑ ML solution

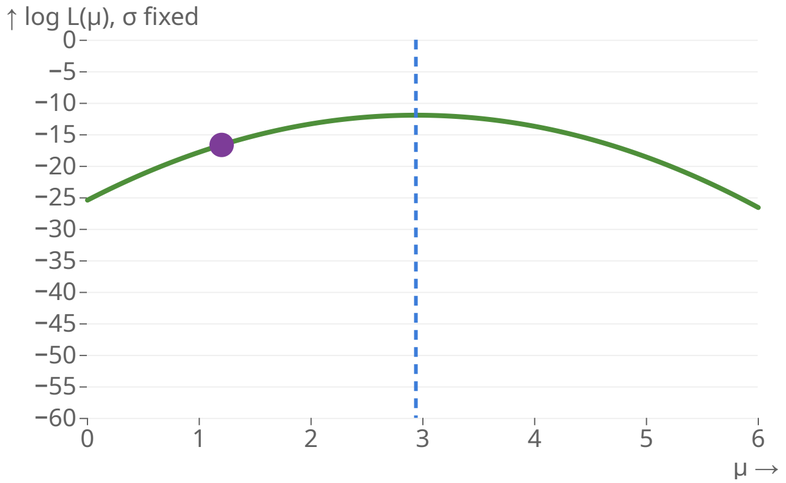
N = 8 samples

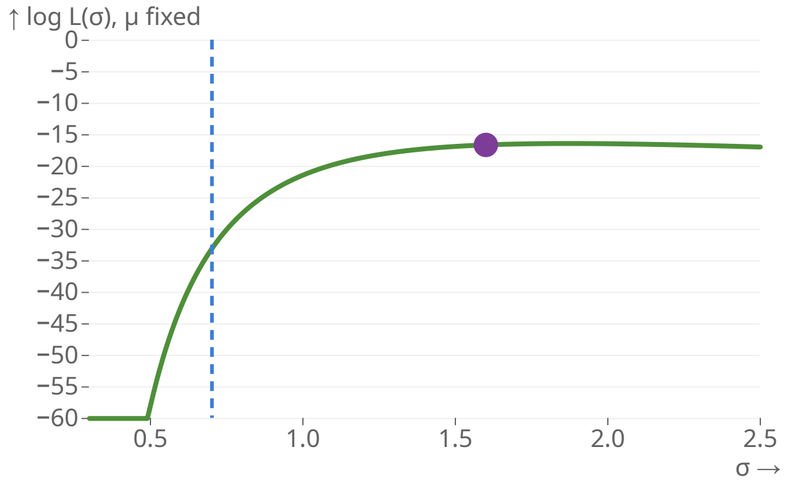
log L(μ, σ) = **-16.60**

ML: μ = 2.94, σ = 0.70

log L = -8.52







Three ways to estimate θ

Maximum likelihood (ML)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

Choose the parameters under which the data are **most probable**:

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} p(T \mid \theta) = \arg\max_{\theta} \prod_{i=1}^{N} p(x_i \mid \theta)$$

Prediction for a new x^*: **plug in** the estimate, $p(x^* \mid T) \approx p(x^* \mid \hat{\theta}_{\mathrm{ML}})$

Result: $\hat{\theta}_{\mathrm{ML}}$, **one number** (a point estimate), not a distribution.

Pros

Simple, no extra assumptions, the best with **a lot of data**.

Cons

With **few** data it trusts them blindly: 3 tosses, 3 heads, so $\hat{\pi}_{\mathrm{head}} = 1$?

Maximum a posteriori (MAP)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

θ is a **random variable** with a **prior** $p(\theta)$. Maximize the **posterior**:

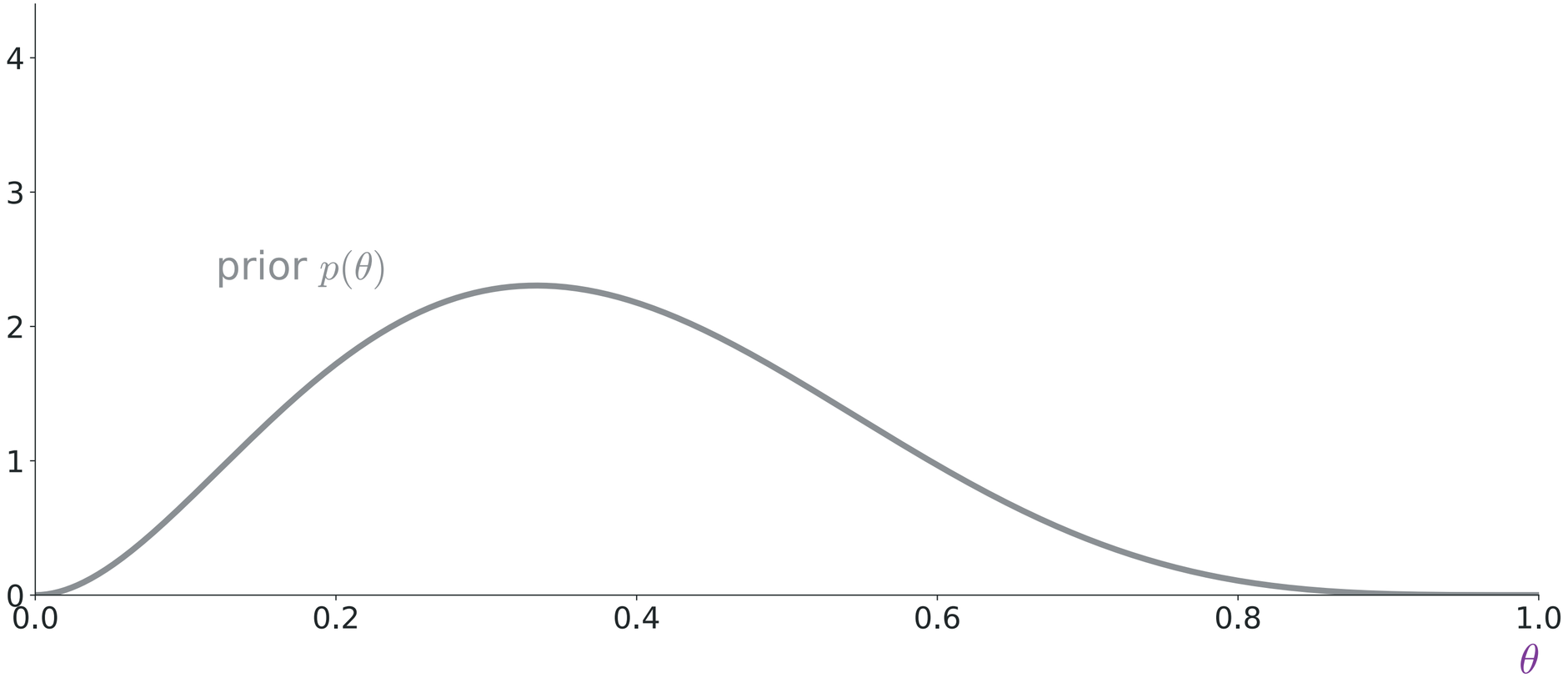
$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid T) = \arg\max_{\theta} \frac{p(T \mid \theta)\, p(\theta)}{p(T)} = \arg\max_{\theta} p(T \mid \theta)\, p(\theta)$$

- $p(T)$ does not depend on θ: drop it
- the prior $p(\theta)$ has to be **specified**

Result: $\hat{\theta}_{\text{MAP}}$, again **one number**.

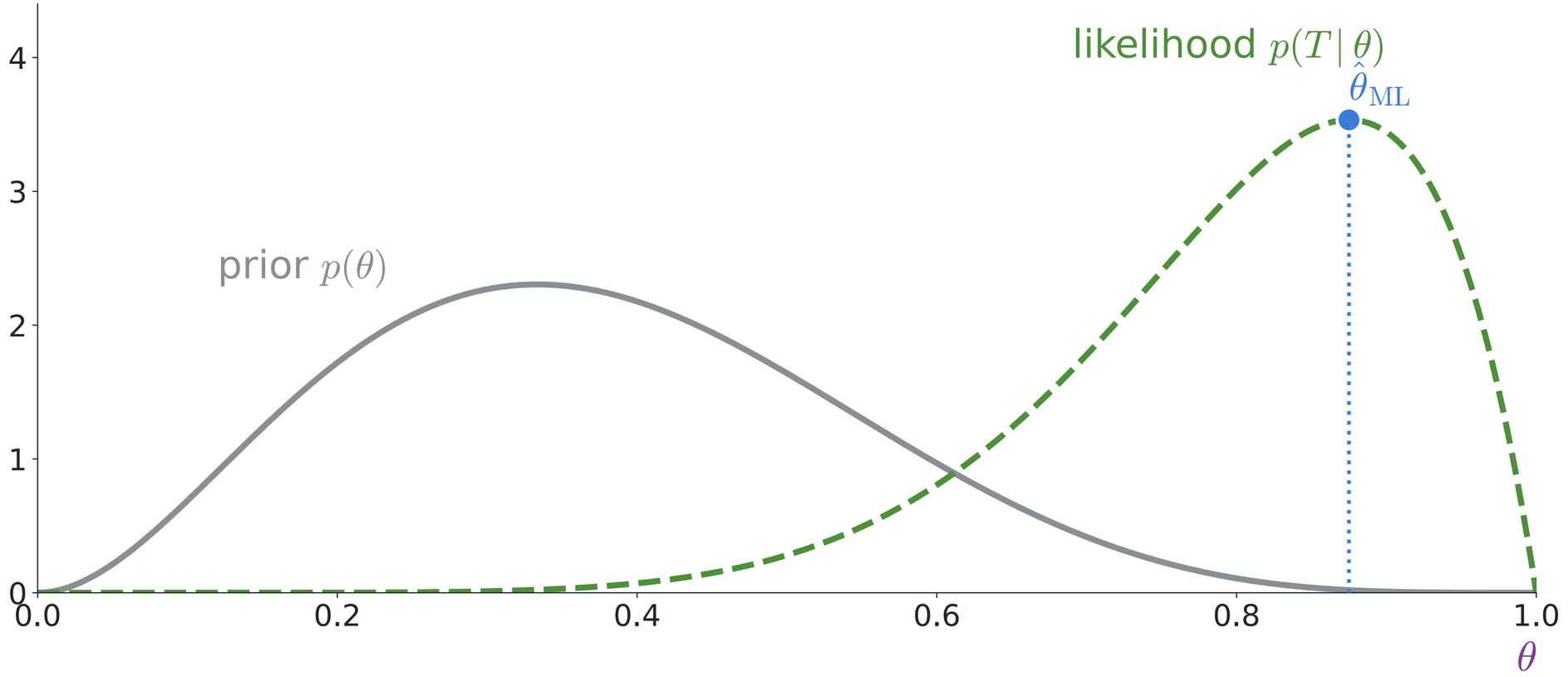
Convenient prior: **conjugate**. The posterior has the same form as the prior.

Prior, likelihood, posterior



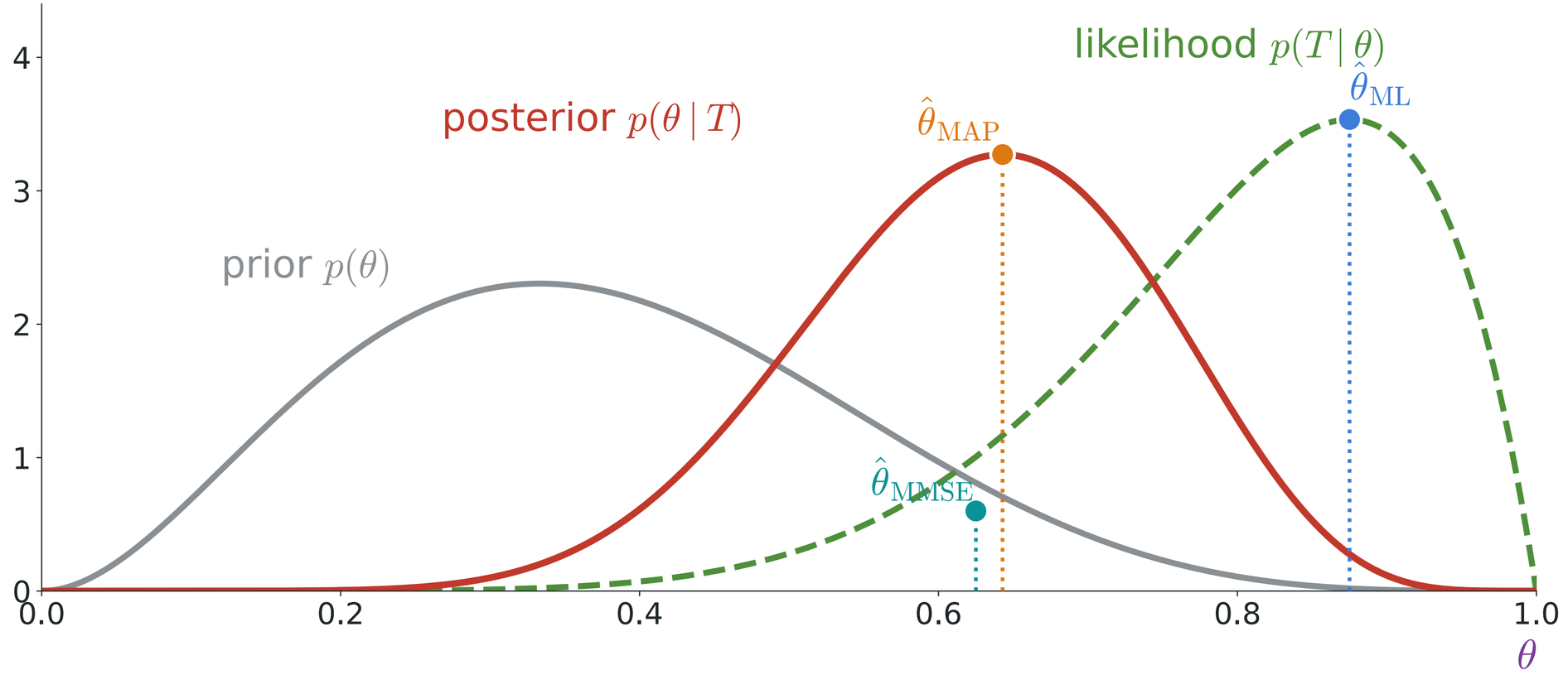
Before the data: the **prior** $p(\theta)$.

Prior, likelihood, posterior



The data: **likelihood** $p(T \mid \theta)$, its maximum is $\hat{\theta}_{\mathrm{ML}}$.

Prior, likelihood, posterior



Posterior $\propto$ likelihood $\times$ prior: a compromise between the two.

Bayesian inference: keep the whole posterior

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

Do not pick one θ, keep the **whole distribution**:

$$p(\theta \mid T) = \frac{p(T \mid \theta)\, p(\theta)}{p(T)}$$

Prediction: **average** all models, weighted by their probability:

$$p(x^* \mid T) = \int p(x^* \mid \theta)\, p(\theta \mid T)\, d\theta$$

Result: **a distribution** $p(\theta \mid T)$, no single θ. The prediction is a distribution too.

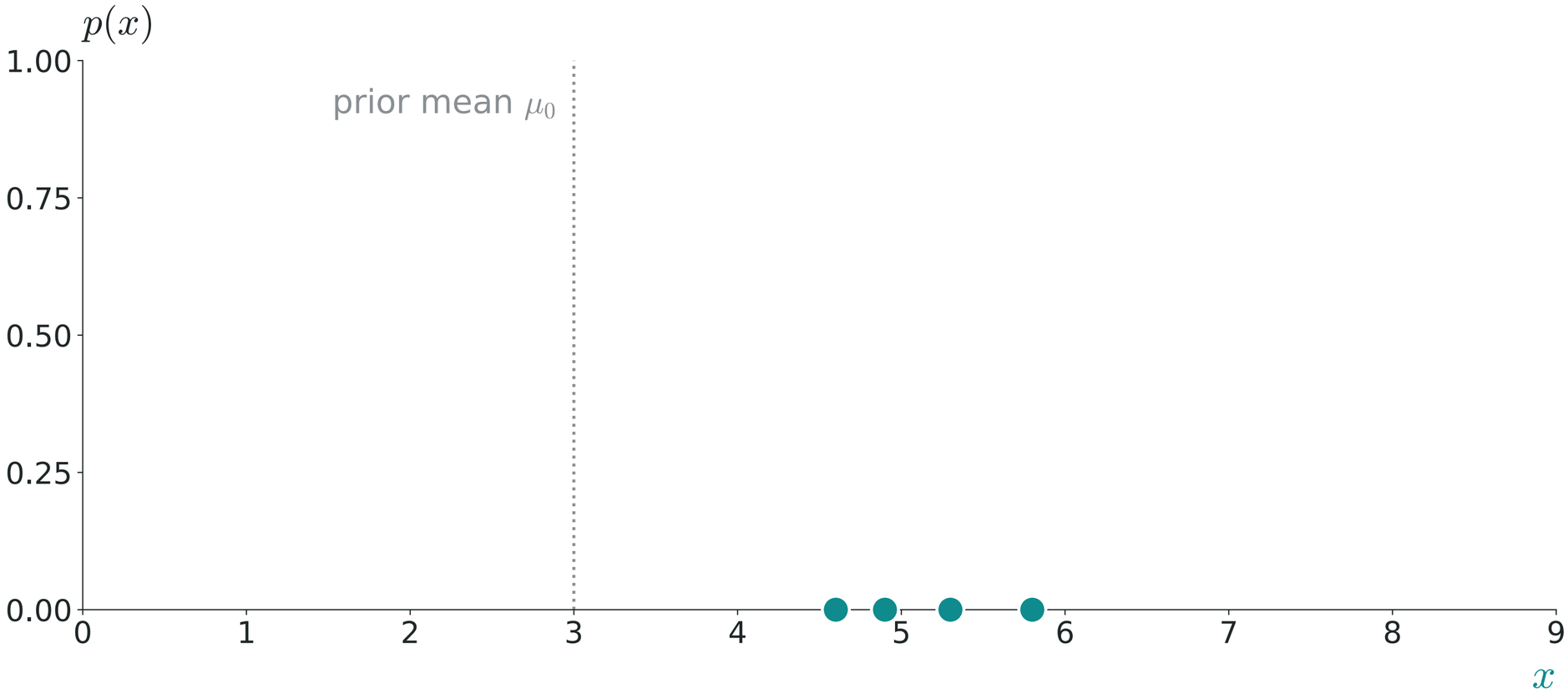
Pros

Uses all the information, knows how **uncertain** it is.

Cons

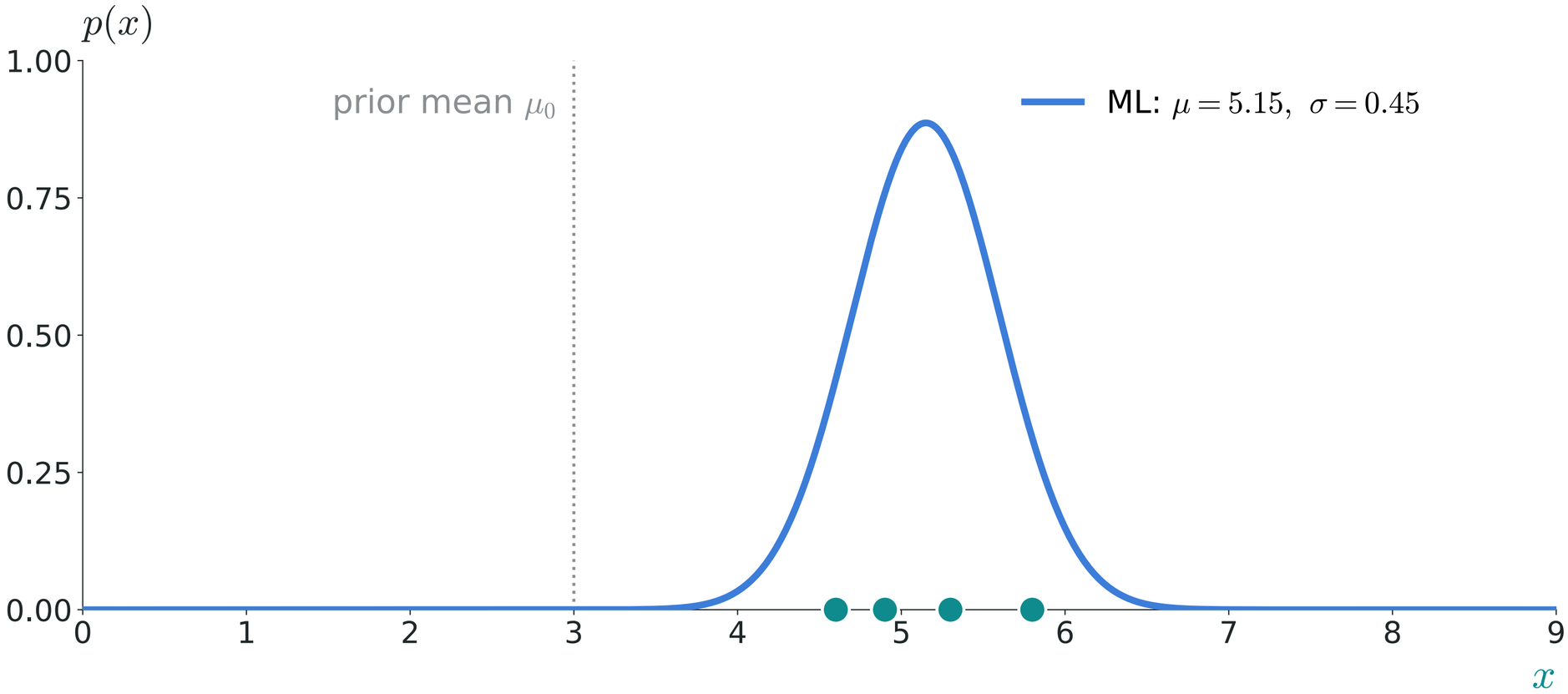
The integral: in closed form only for **conjugate** priors.

Comparing the estimates



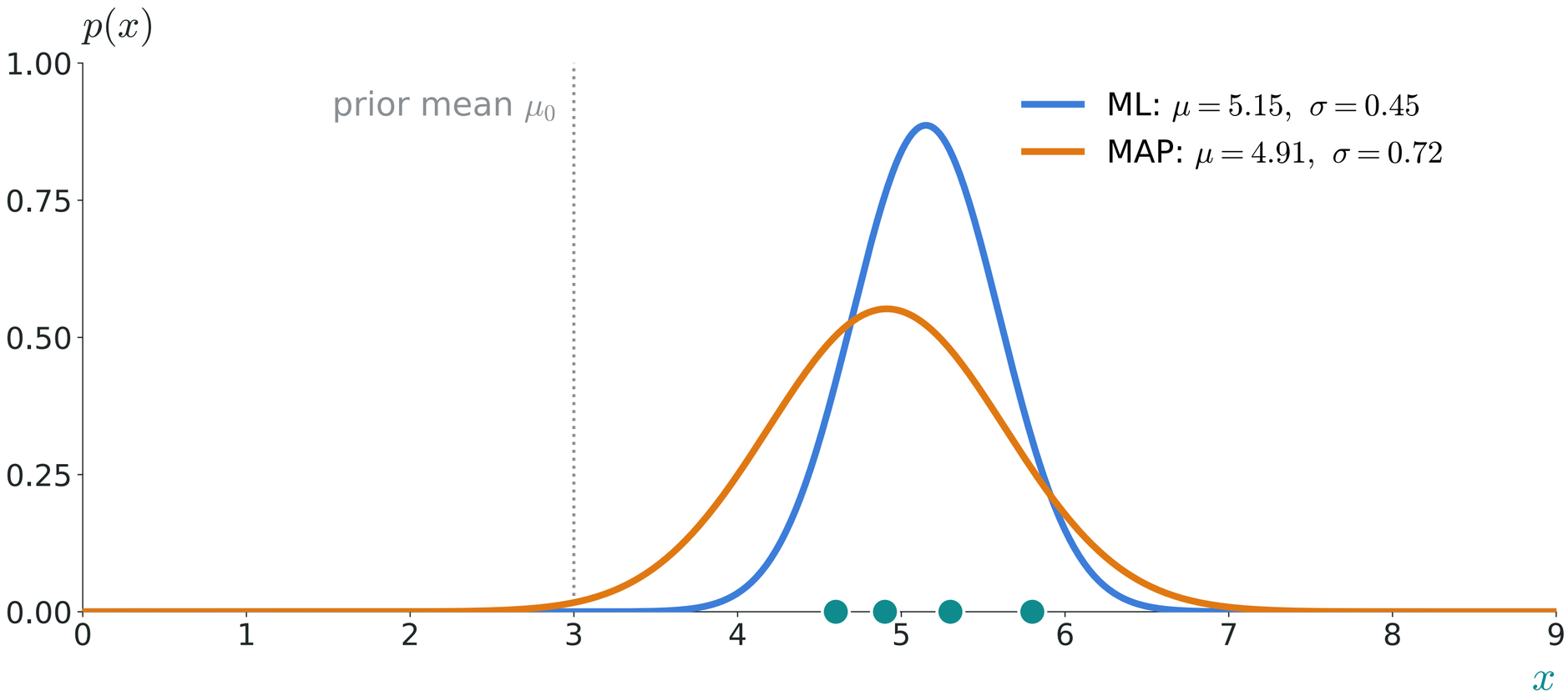
$N = 4$ samples, normal model, unknown μ and σ^2. The prior puts μ around $\mu_0 = 3$.

Comparing the estimates



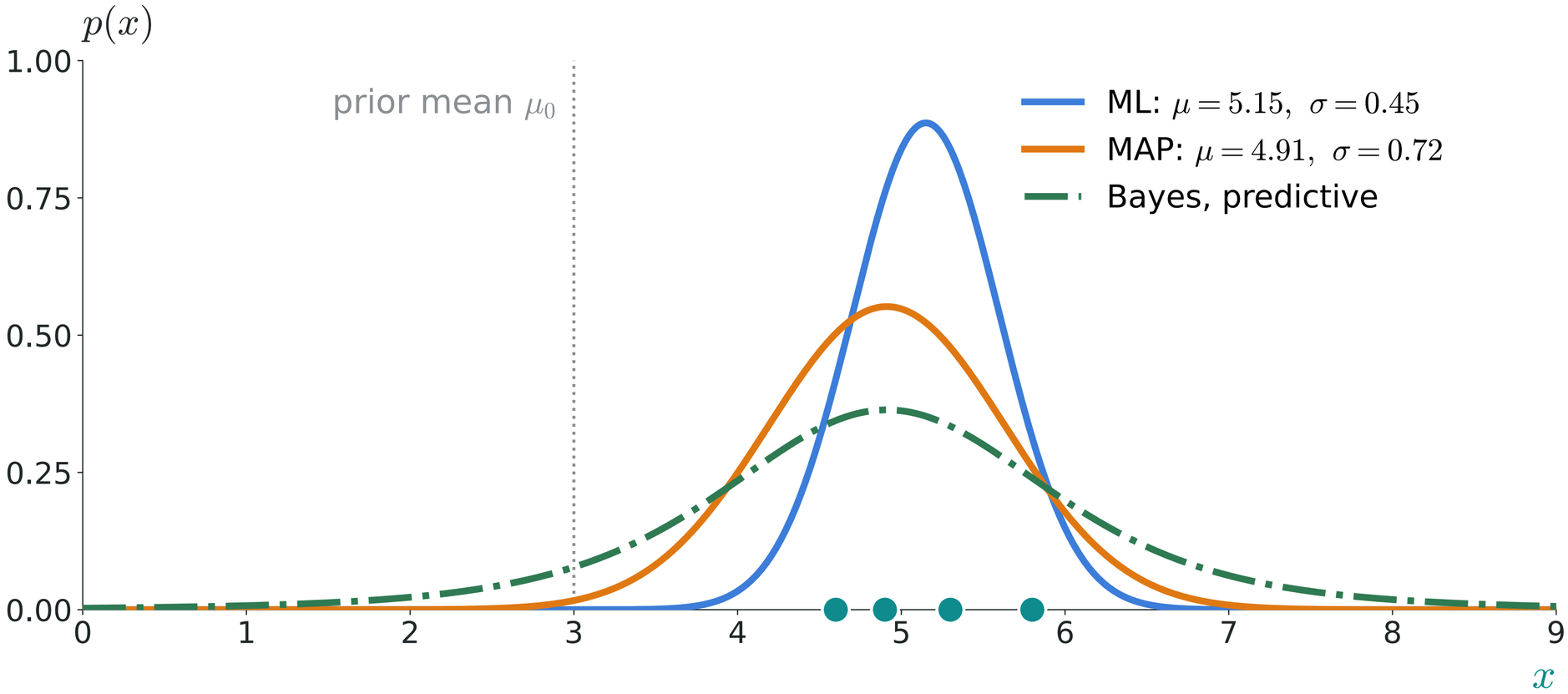
ML: plug in $\hat{\theta}_{\mathrm{ML}}$. It fits the 4 samples tightly, too narrow.

Comparing the estimates



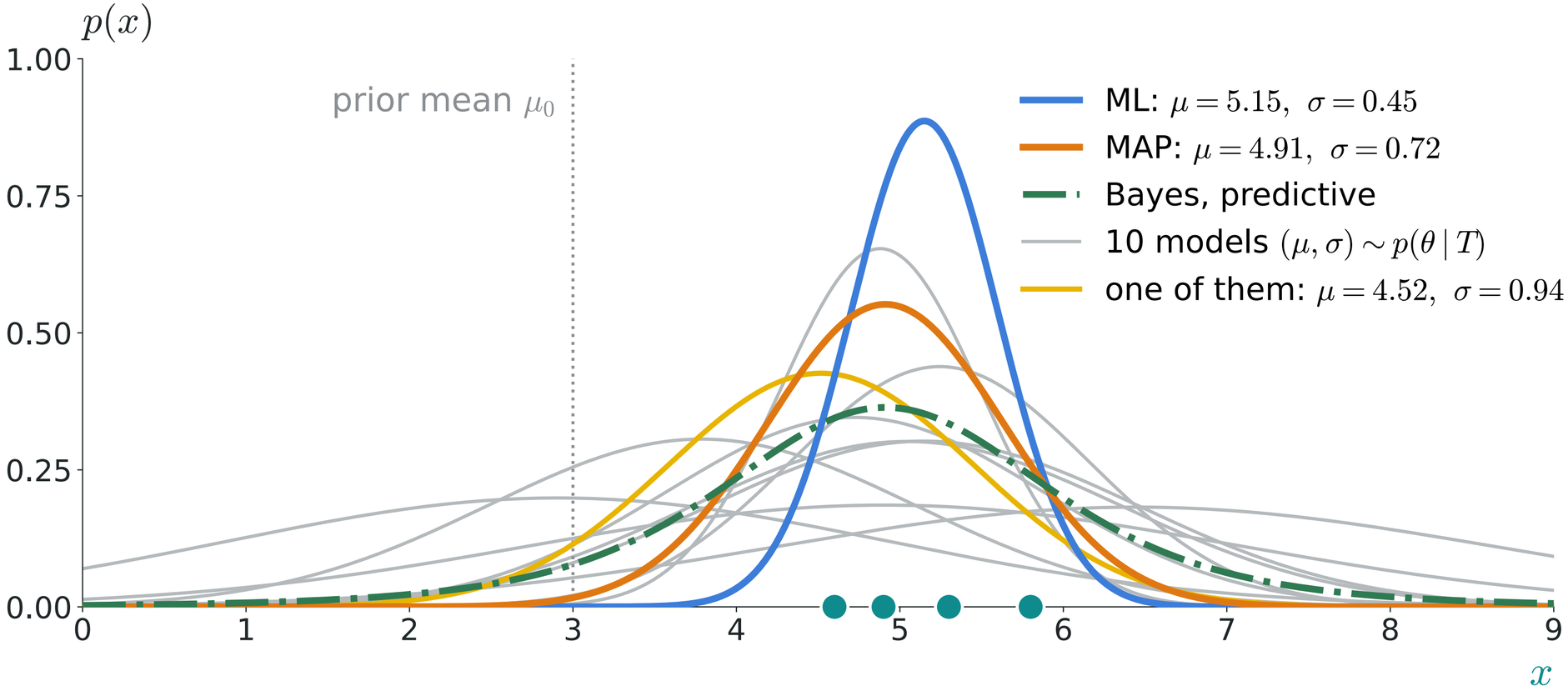
MAP: the prior pulls μ towards μ_0 and makes σ larger.

Comparing the estimates



Bayes: the predictive $p(x^* \mid T)$, not a normal any more.

Comparing the estimates



It averages models from $p(\theta \mid T)$: 10 of them drawn, **one** highlighted.

Bayesian parameter estimation (MMSE)

$T = \{x_1, \ldots, x_N\}$ training set, x_i **i.i.d.** (independent, identically distributed) $\Rightarrow$ $p(T \mid \theta) = \prod_{i=1}^{N} p(x_i \mid \theta)$

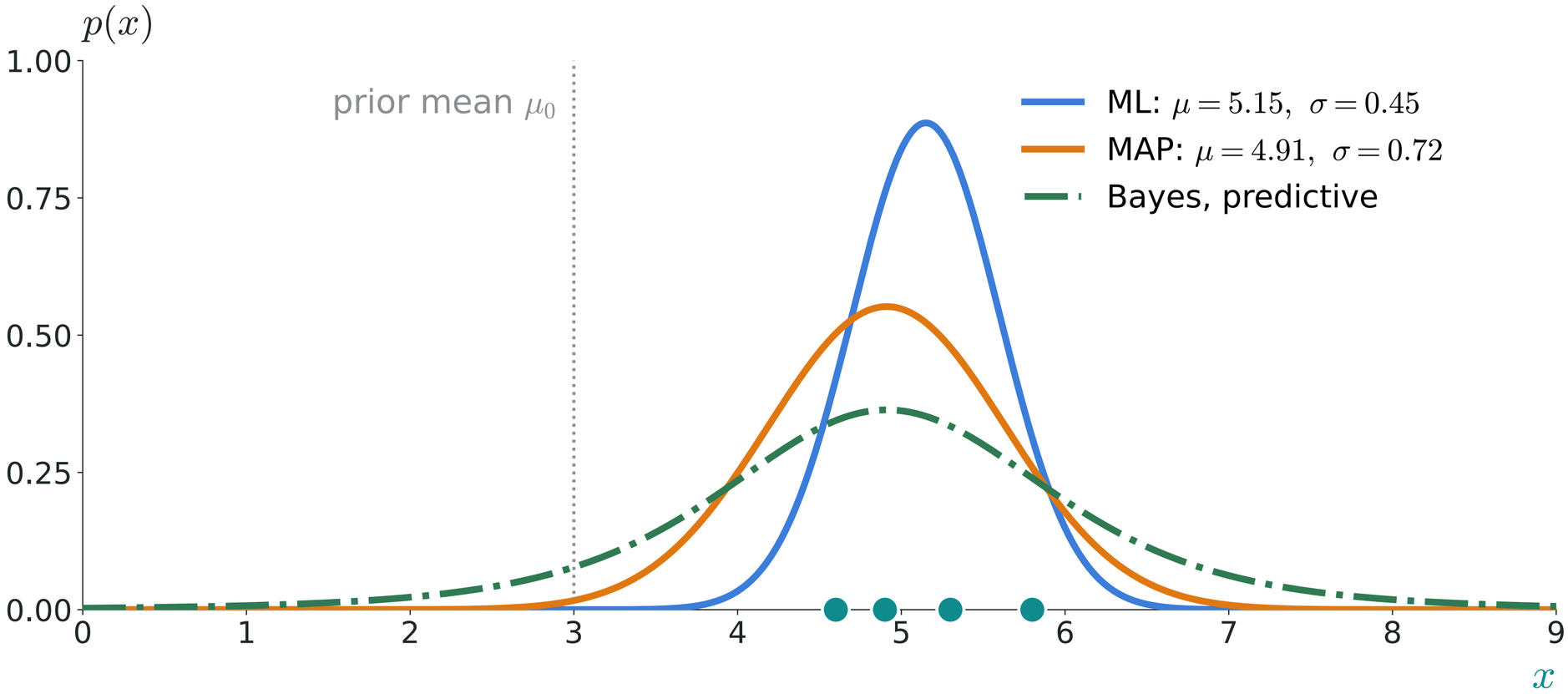
Or: pick **one** $\hat{\theta}$ with the smallest expected **squared error**:

$$R(\hat{\theta}) = \int p(\theta \mid T)\,(\theta - \hat{\theta})^2\, d\theta$$

$$\frac{\partial R}{\partial \hat{\theta}} = -2 \int p(\theta \mid T)\,(\theta - \hat{\theta})\, d\theta = 0 \quad \Rightarrow \quad \hat{\theta}_{\mathrm{MMSE}} = \int \theta\, p(\theta \mid T)\, d\theta$$

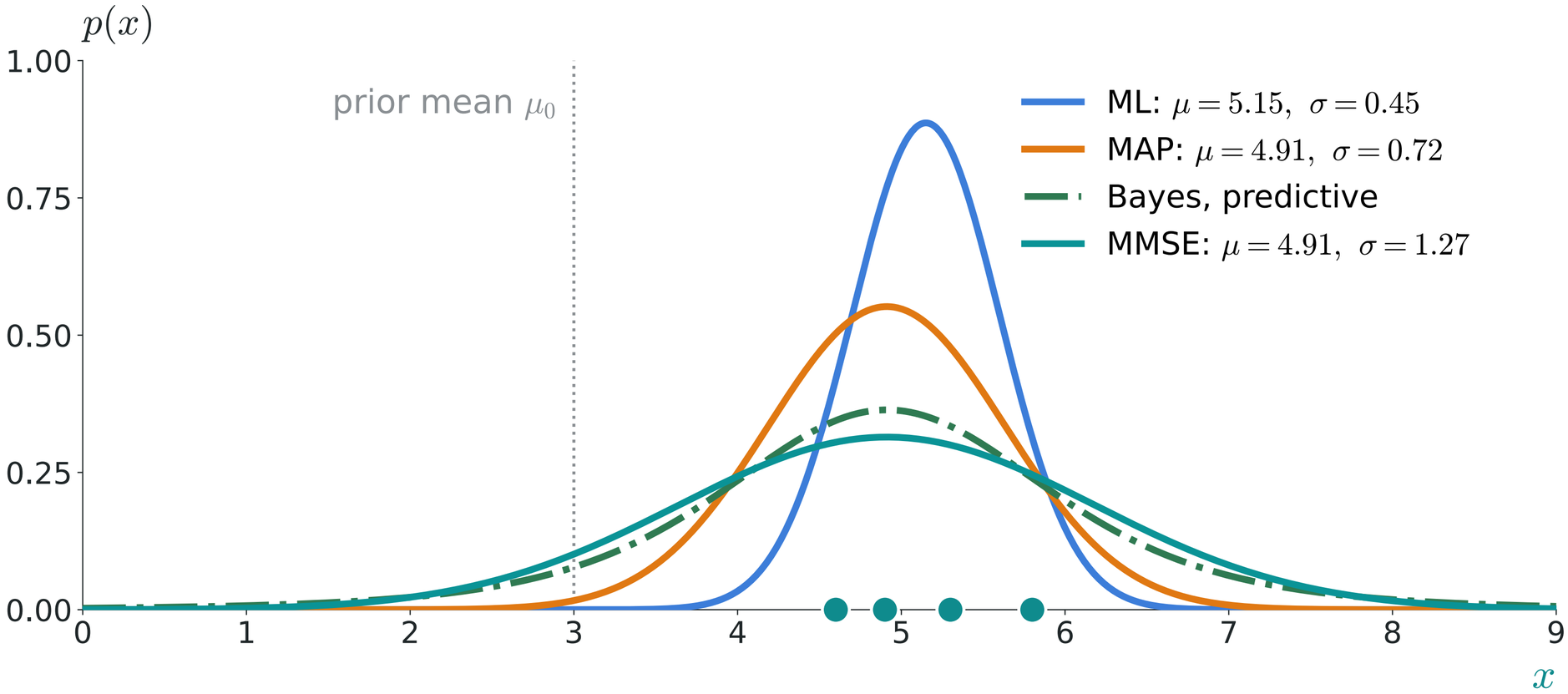
Result: $\hat{\theta}_{\mathrm{MMSE}} = \mathbb{E}[\theta \mid T]$, **one number**: the mean of the posterior. Prediction by plugging in.

Comparing the estimates: MMSE



ML, MAP and Bayes from before.

Comparing the estimates: MMSE



MMSE: plug in the posterior mean. Still a normal, unlike **Bayes**.

Three estimates, one picture

	prior	result	prediction $p(x^* \mid T)$
ML	no	$\hat{\theta}_{\mathrm{ML}}$, max of likelihood	$p(x^* \mid \hat{\theta}_{\mathrm{ML}})$
MAP	yes	$\hat{\theta}_{\mathrm{MAP}}$, max of posterior	$p(x^* \mid \hat{\theta}_{\mathrm{MAP}})$
MMSE	yes	$\hat{\theta}_{\mathrm{MMSE}}$, mean of posterior	$p(x^* \mid \hat{\theta}_{\mathrm{MMSE}})$
Bayes	yes	the whole $p(\theta \mid T)$	$\int p(x^* \mid \theta)\, p(\theta \mid T)\, d\theta$

With a **flat** prior, MAP equals ML.

With **a lot of data**, all four agree.

Conjugate priors

likelihood $p(x \mid \theta)$	parameter	conjugate prior	posterior
binomial	π	Beta (a, b)	Beta $(a + R,\ b + N - R)$
categorical	$p_1, \ldots, p_K$	Dirichlet $(\alpha_1, \ldots, \alpha_K)$	Dirichlet $(\alpha_k + N_k)$
normal	μ (σ^2 known)	normal (μ_0, σ_0^2)	normal
normal	μ, σ^2	NIG	NIG assignment
Poisson	λ	Gamma (α, β)	Gamma $(\alpha + \sum x_i,\ \beta + N)$
uniform $[0, \theta]$	θ	Pareto (θ_0, k)	Pareto $(\max(\theta_0, m),\ k + N)$

The full table: Wikipedia, Conjugate prior.

The German tank problem

Counting tanks from serial numbers



WW2: the Allies read **serial numbers** on captured tanks (gearboxes, wheels). How many tanks does Germany produce per month?

month	intelligence	statistics	German records
June 1940	1000	169	**122**
June 1941	1550	244	**271**
August 1942	1550	327	**342**

Ruggles and Brodie (1947); table from Wikipedia, *German tank problem*.

? We have seen n tanks, the largest serial number is m. How many tanks are there?

The model: uniform distribution

Serial numbers are uniform on $[0, \theta]$, θ is the **number of tanks**:

$$p(x \mid \theta) = \begin{cases} \frac{1}{\theta} & 0 \le x \le \theta \\ 0 & \text{otherwise} \end{cases}$$

Continuous for simplicity, real serial numbers are integers.

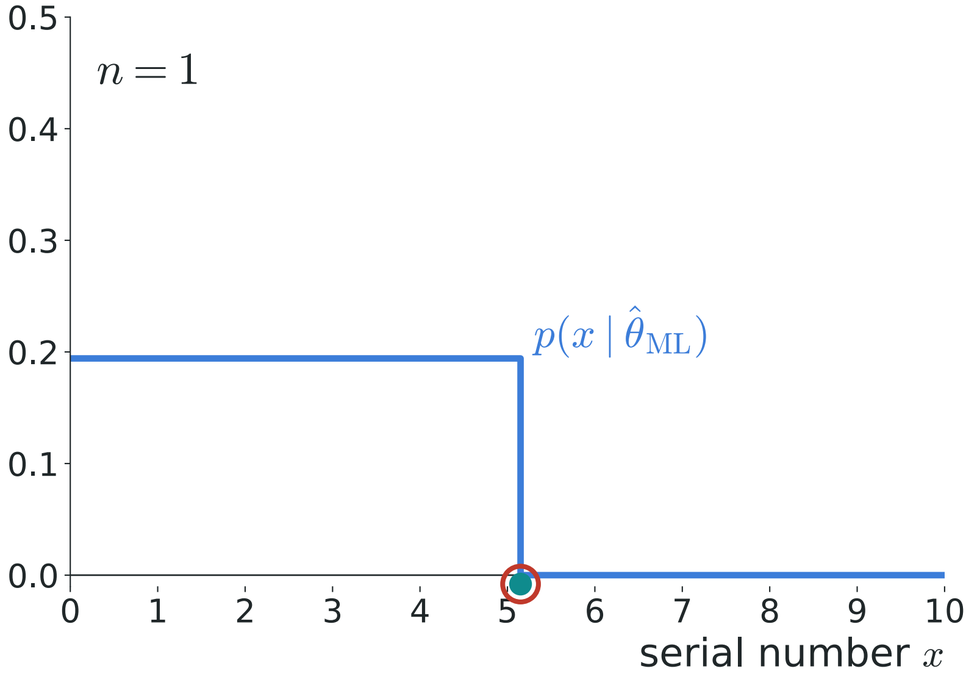
$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} [\![x_i \le \theta]\!]$$

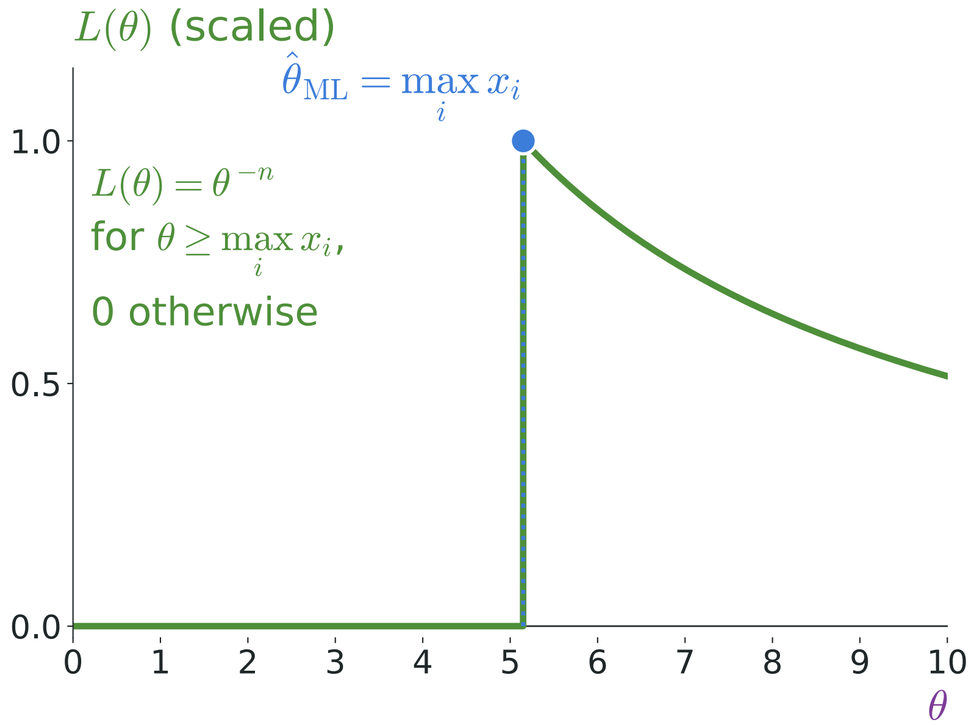
$$L(\theta) = \begin{cases} \theta^{-n} & \theta \ge \max_i x_i \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{\theta}_{\text{ML}} = \max_i x_i = m$$

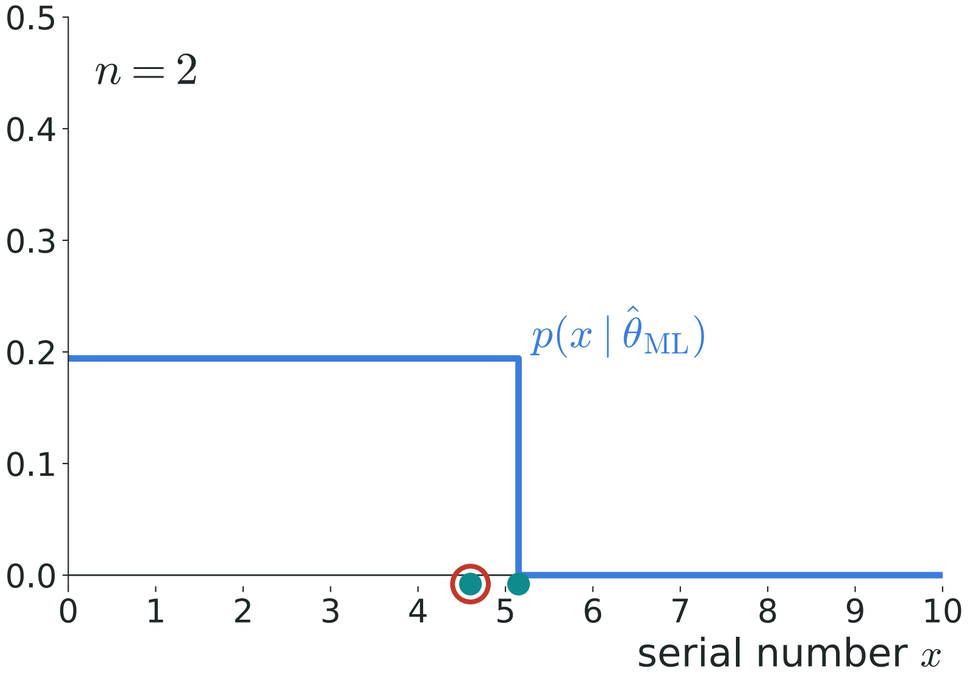
? Is it a good estimate? Can the true θ be **smaller** than m?

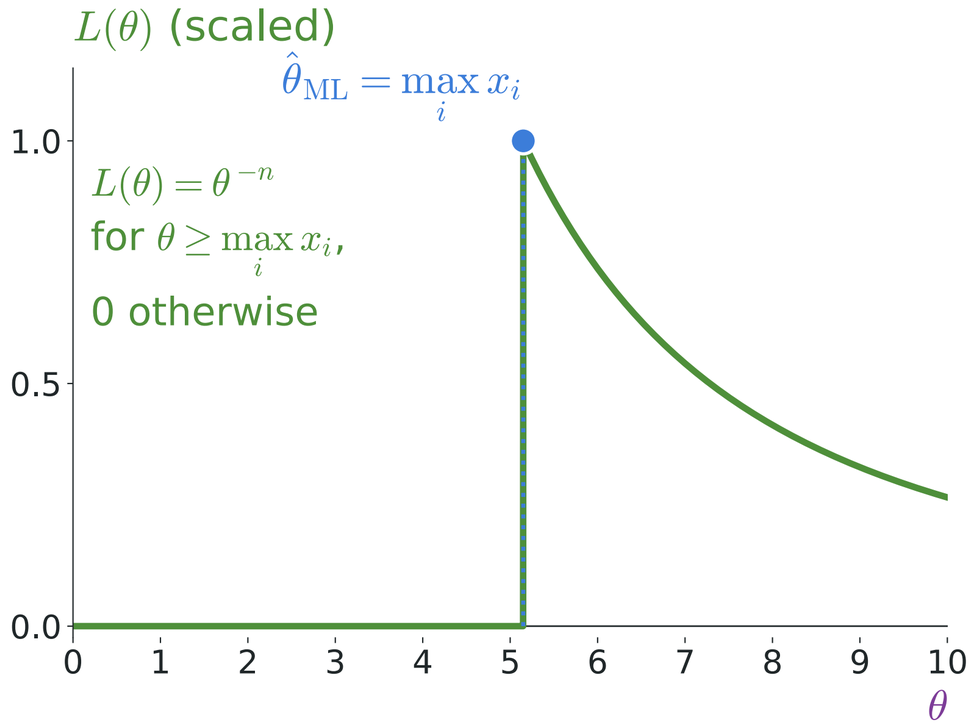
ML with more and more tanks



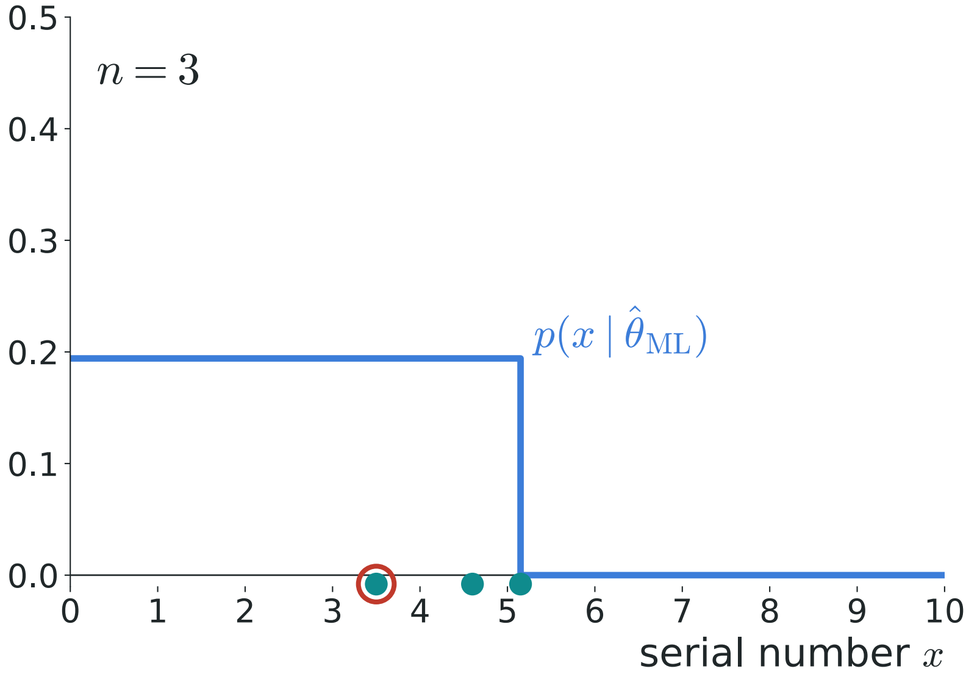


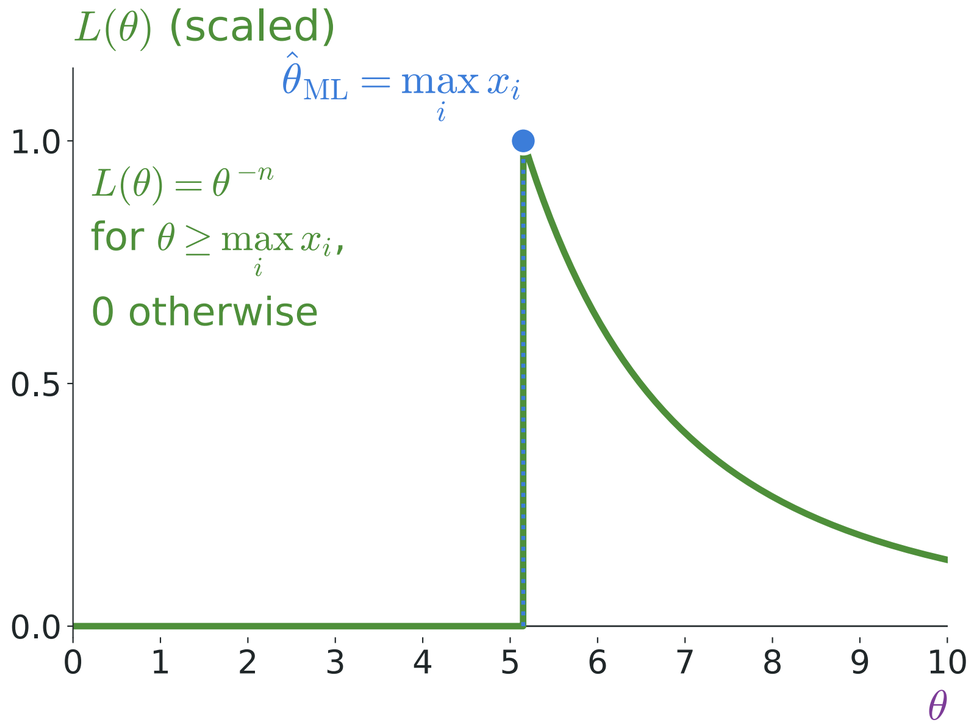
ML with more and more tanks



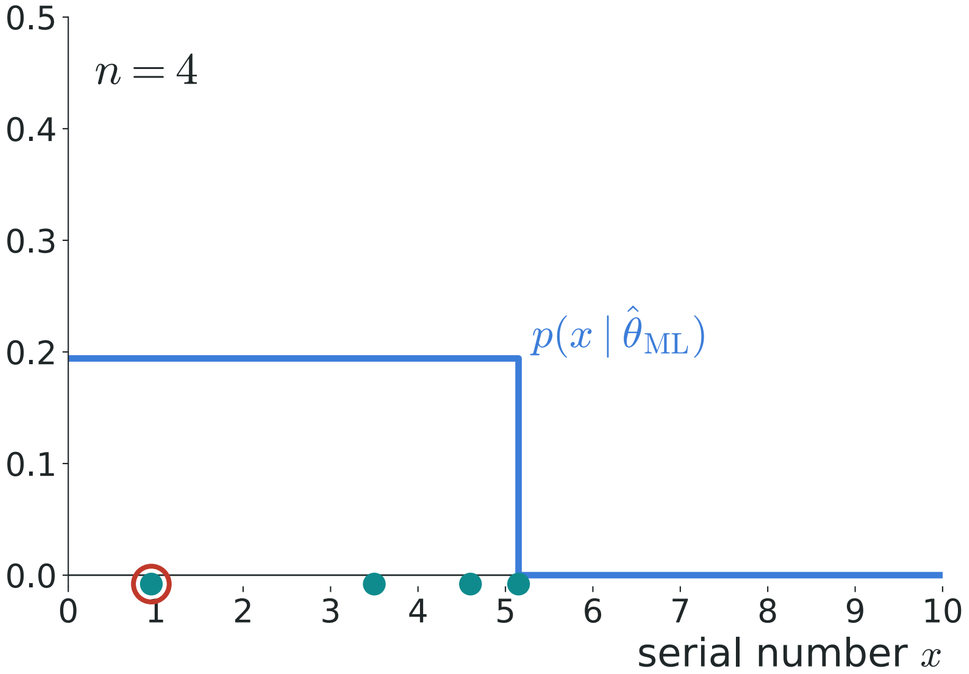


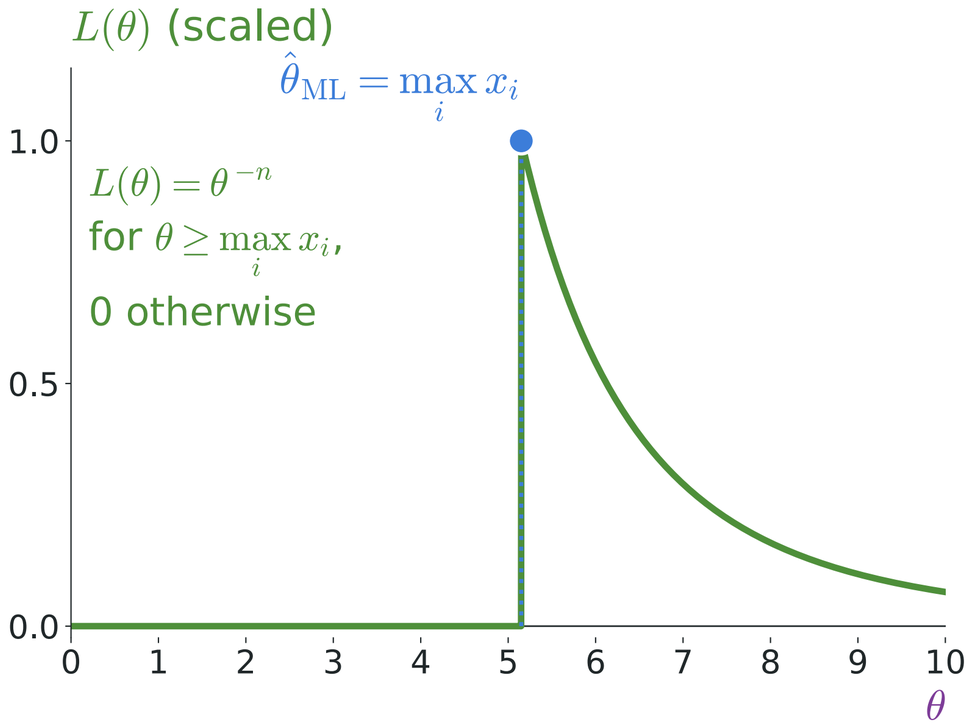
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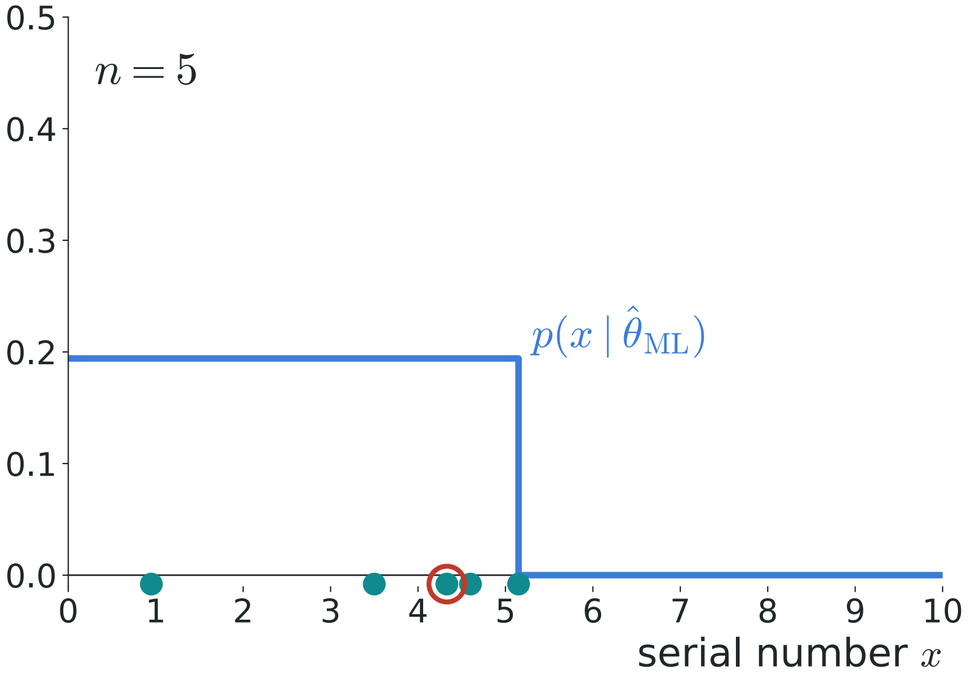


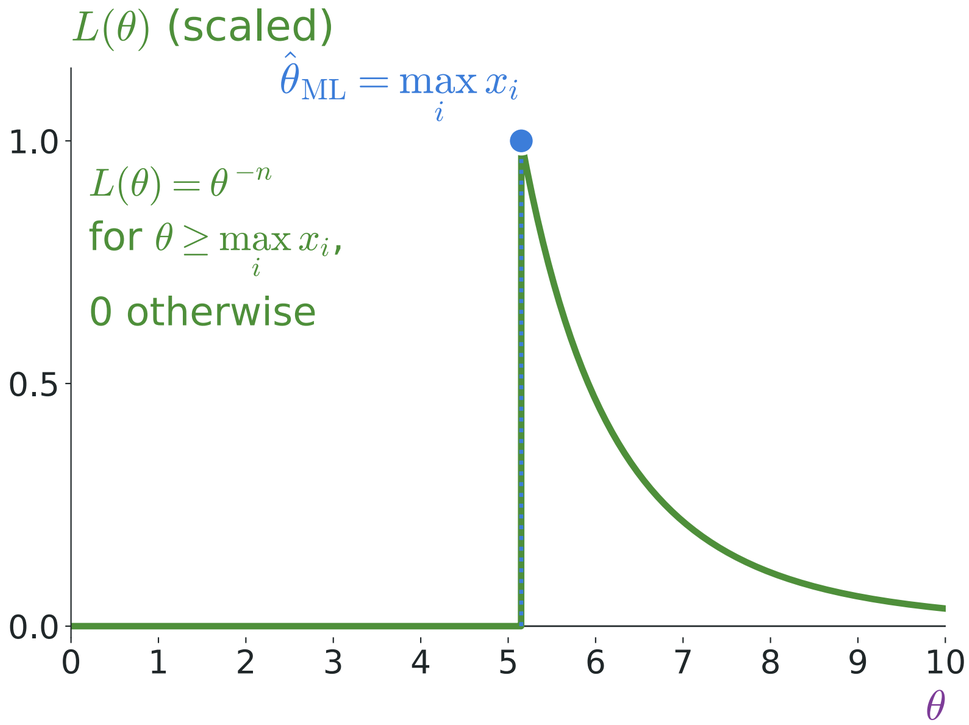
ML with more and more tanks



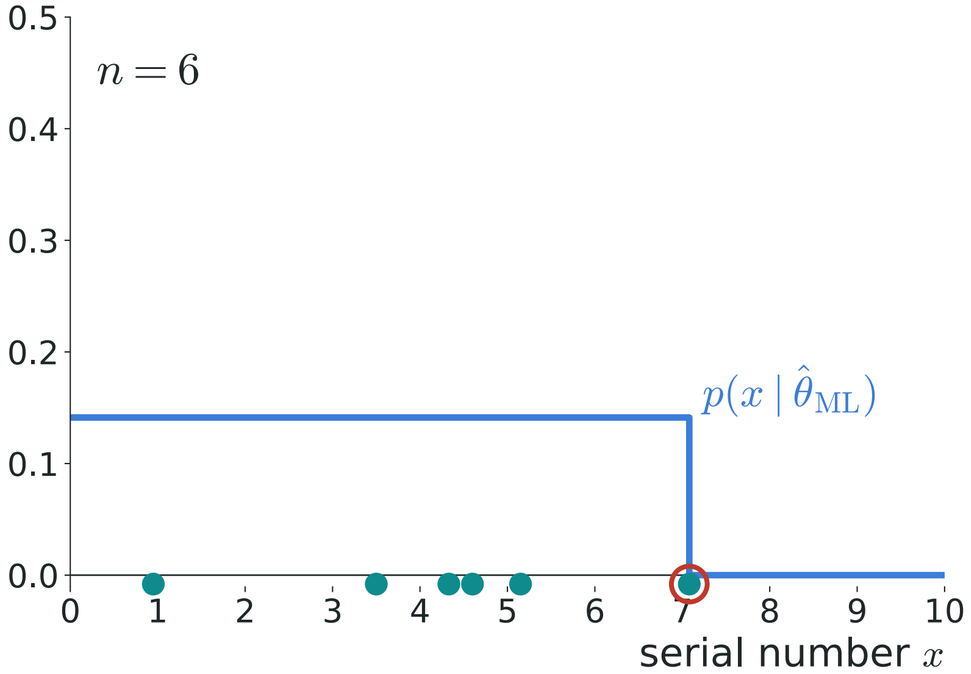


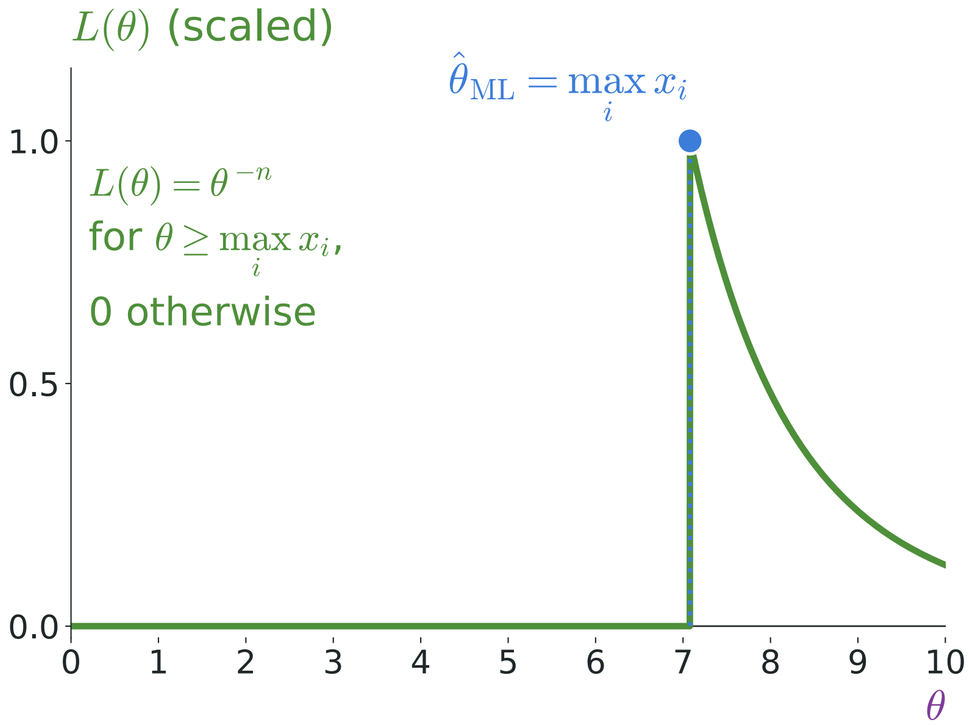
ML with more and more tanks



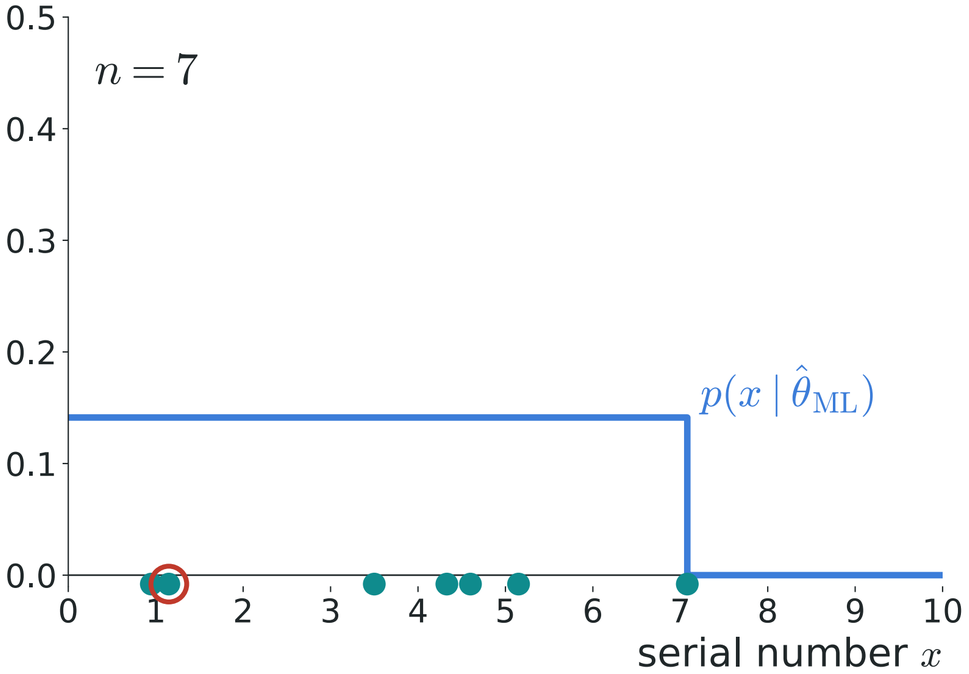


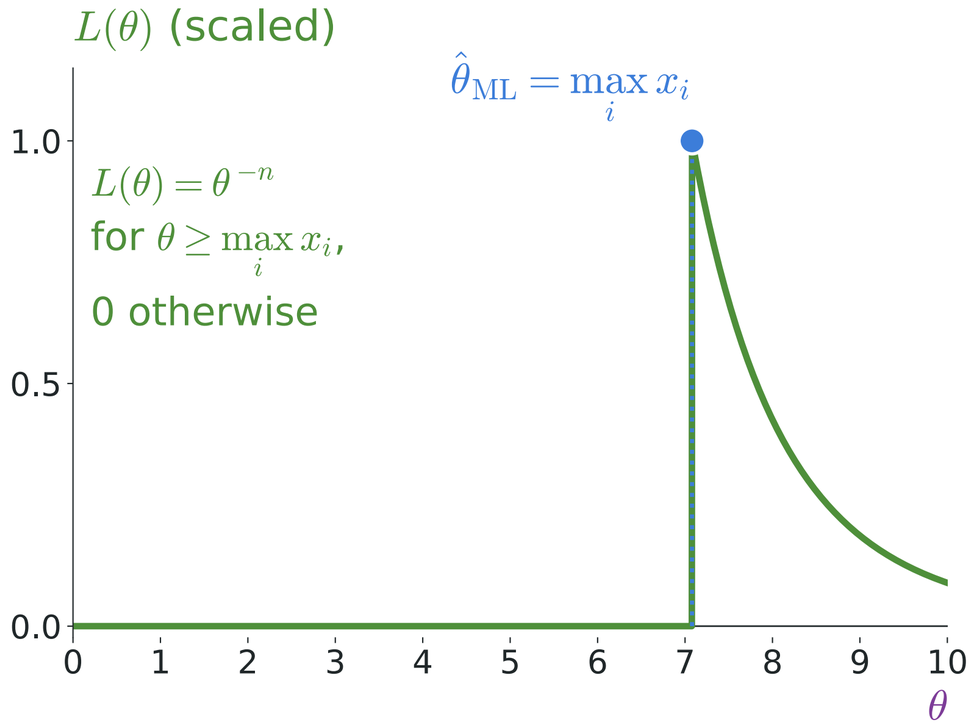
ML with more and more tanks



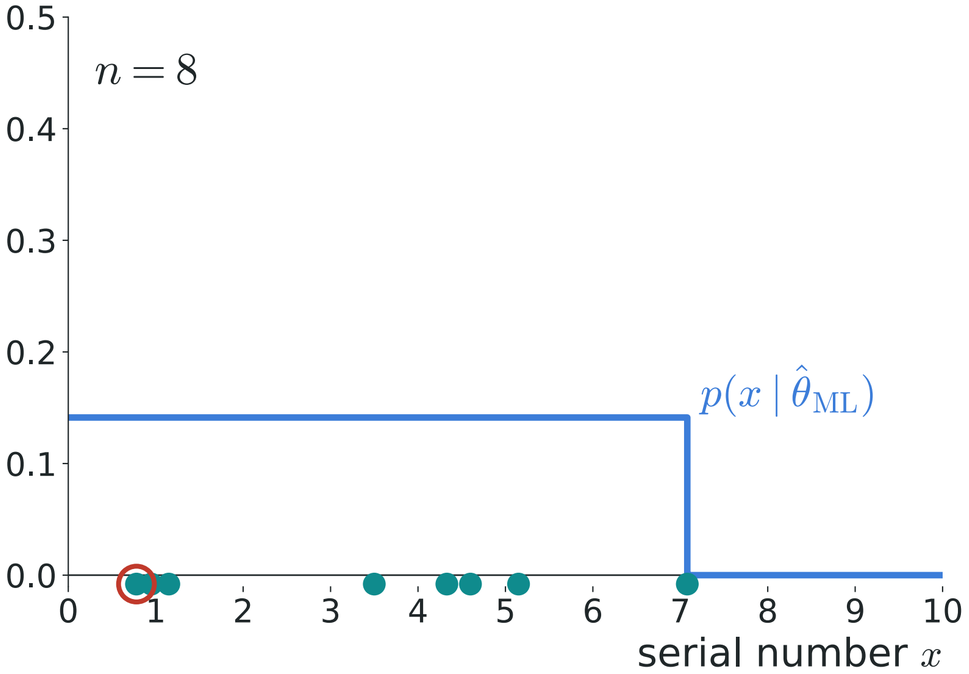


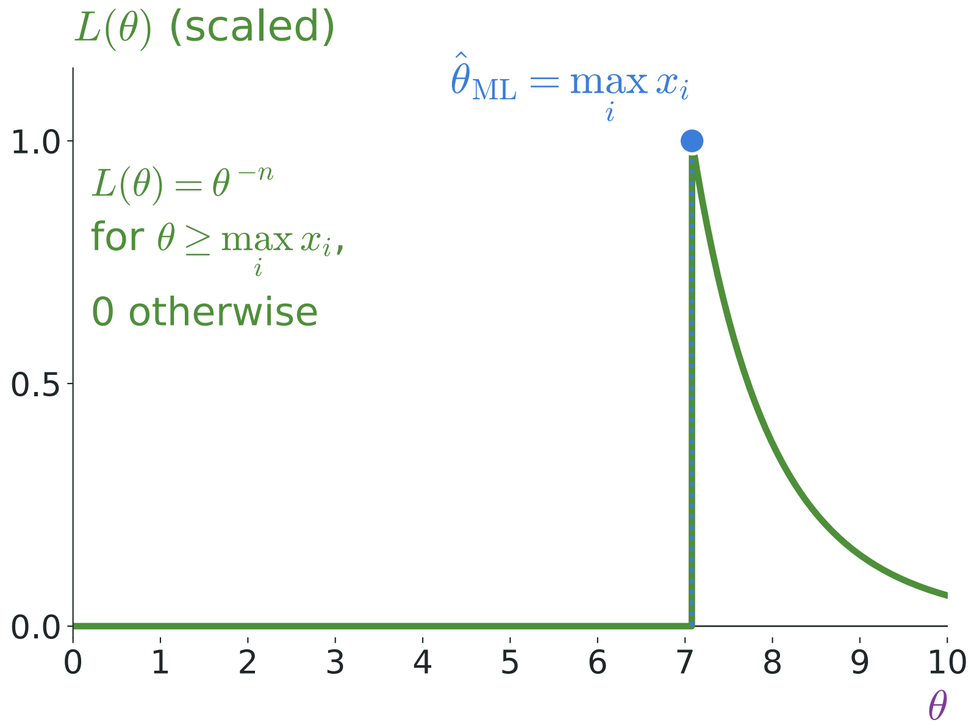
ML with more and more tanks



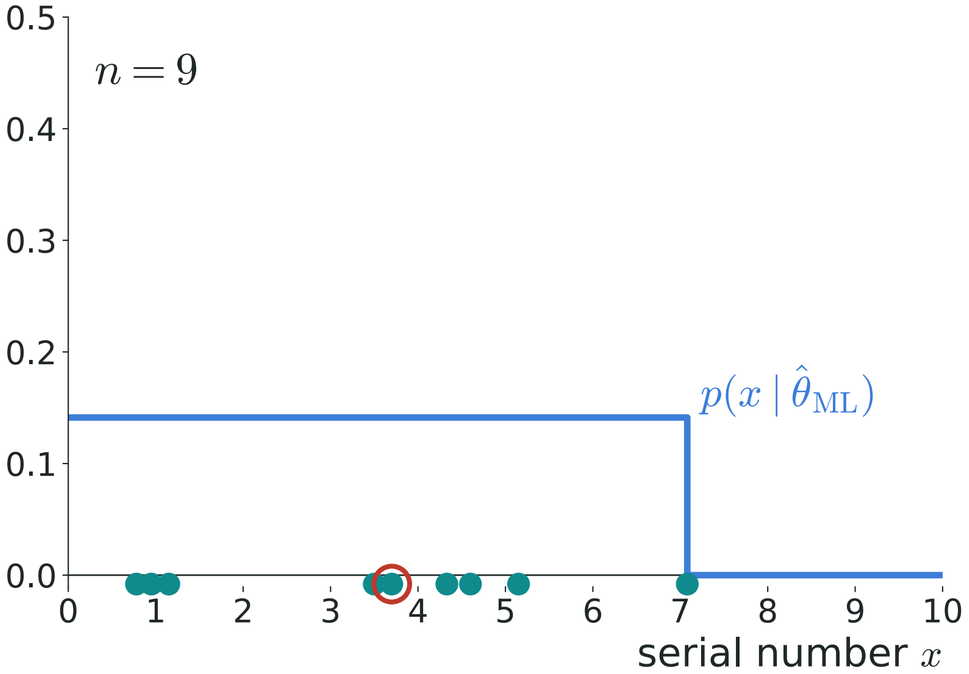


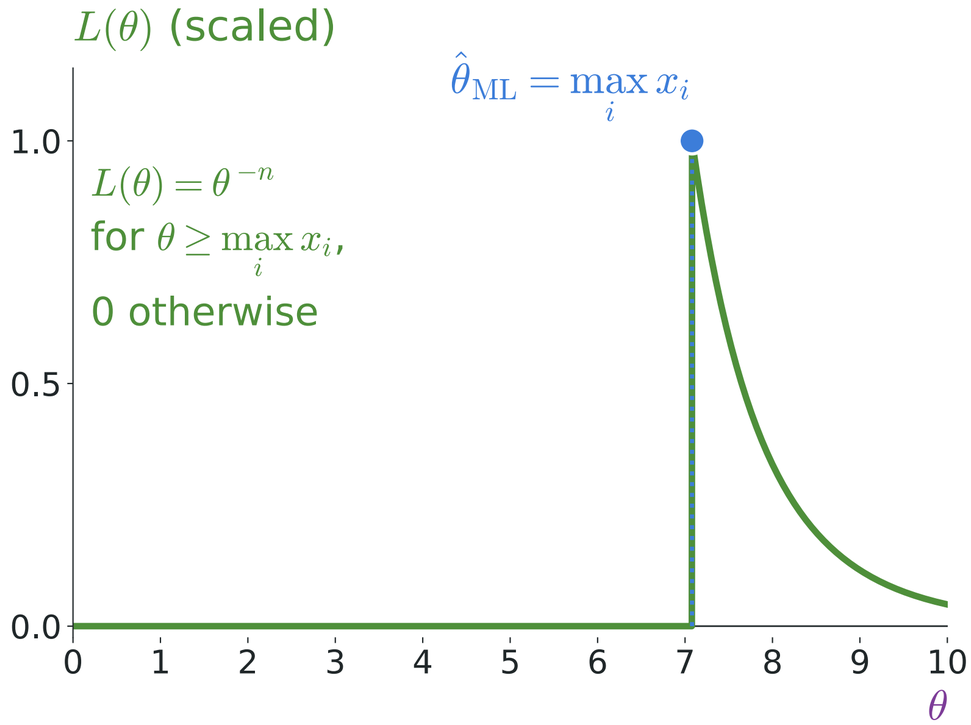
ML with more and more tanks



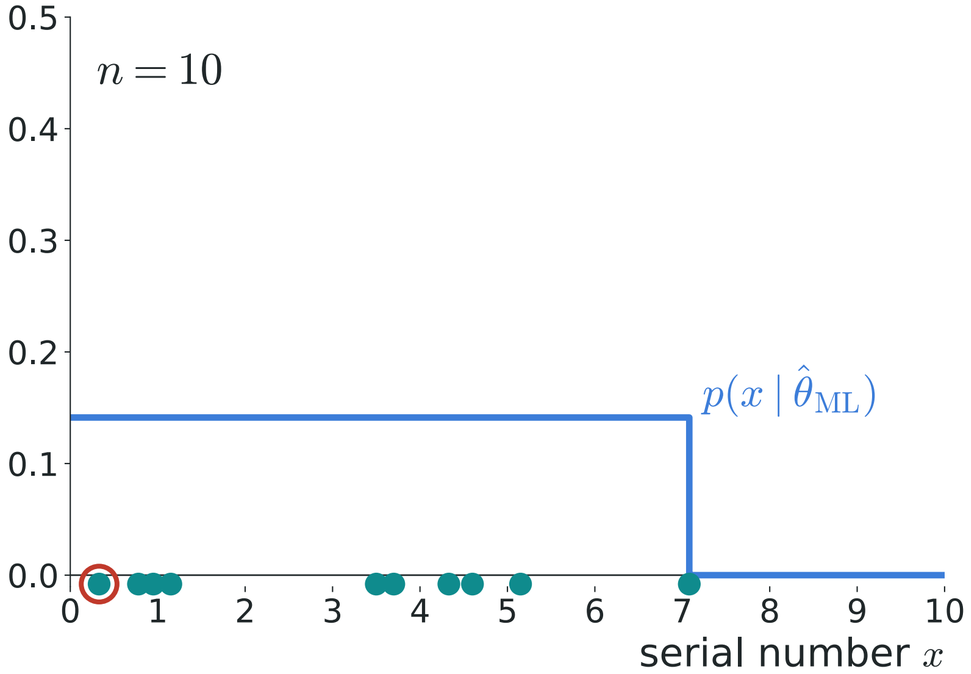


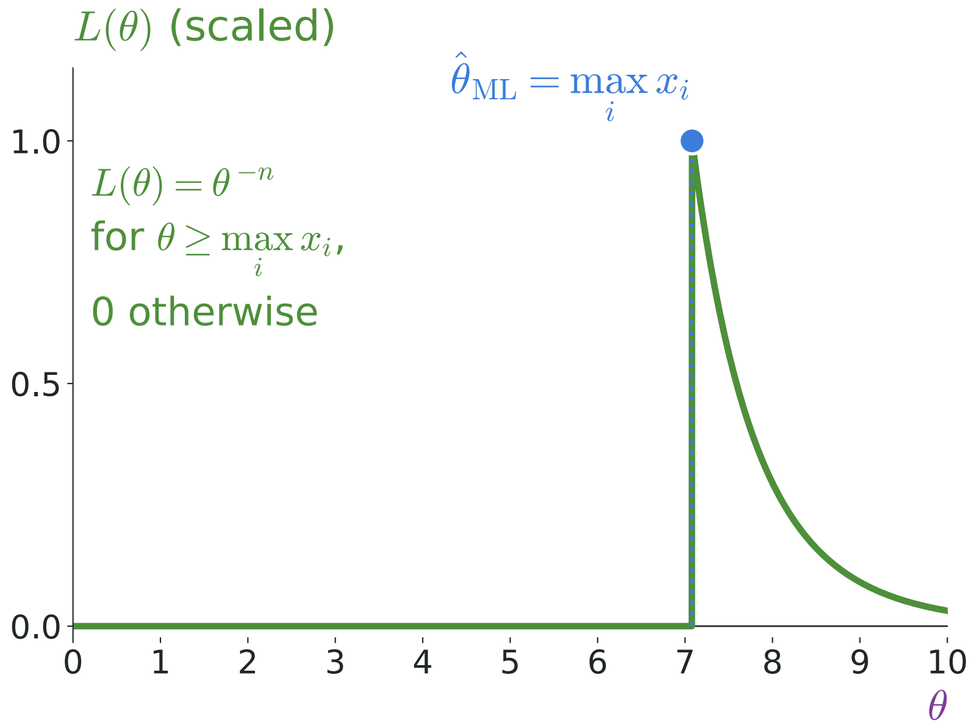
ML with more and more tanks





ML with more and more tanks





Derivation: Bayes for the tanks

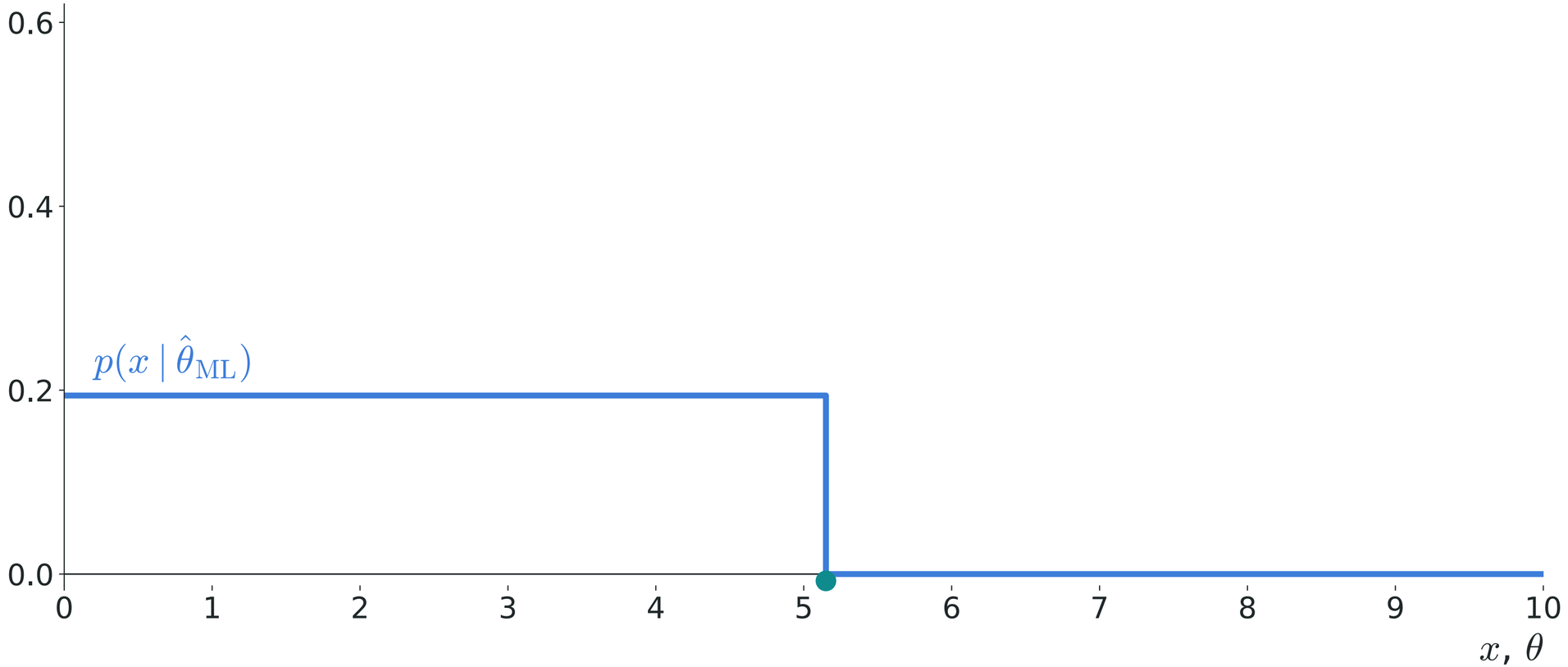
Prior: θ uniform on $[0, M]$, at most M tanks. Posterior, $m = \max_i x_i$:

$$p(\theta \mid T) \propto \prod_{i=1}^{n} \frac{1}{\theta} [\![x_i \le \theta]\!] \, \frac{1}{M} [\![\theta \le M]\!] \quad \Rightarrow \quad p(\theta \mid T) = \frac{1}{Z} \frac{1}{\theta^n} [\![m \le \theta \le M]\!]$$

$$Z = \int_m^M \frac{d\theta}{\theta^n} = \frac{1}{n-1} \left(\frac{1}{m^{n-1}} - \frac{1}{M^{n-1}} \right), \qquad \hat{\theta}_{\mathrm{MAP}} = m = \hat{\theta}_{\mathrm{ML}} \ \ \text{(flat prior)}$$

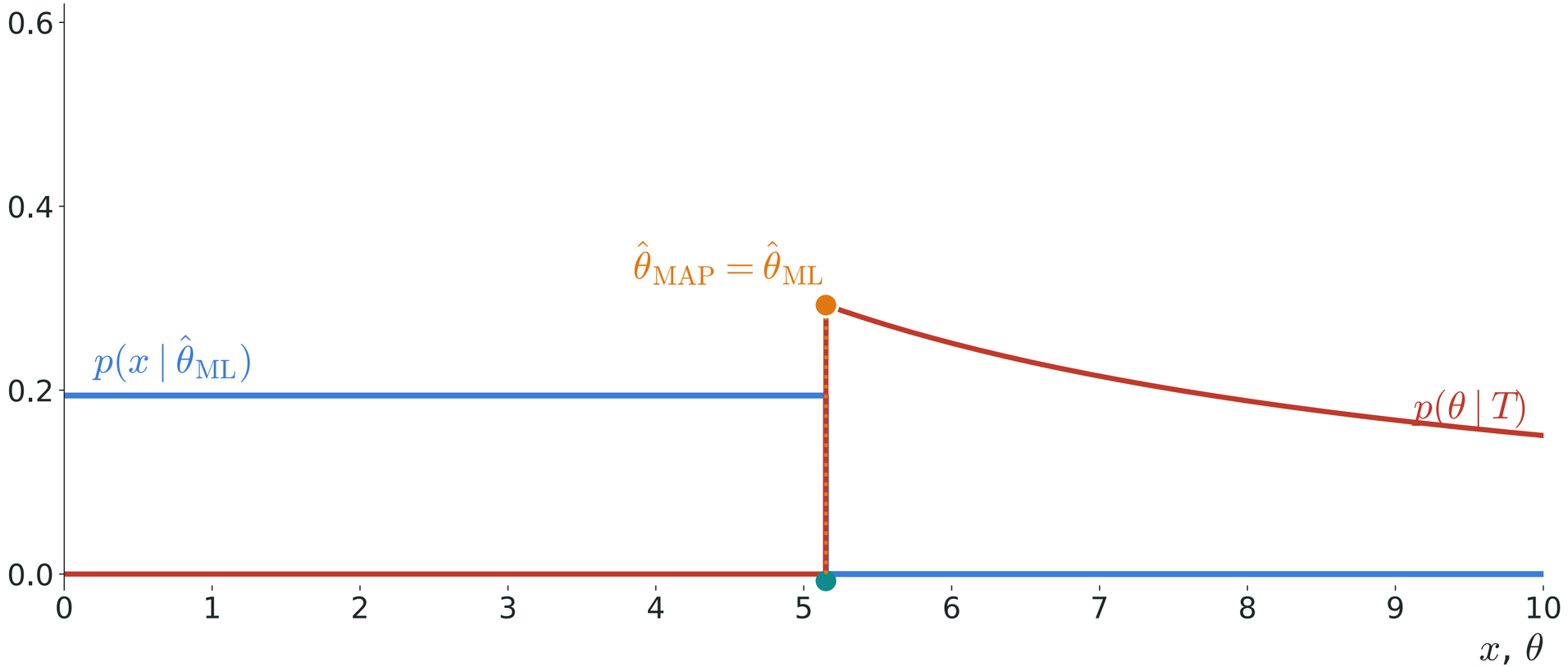
$$\hat{\theta}_{\mathrm{MMSE}} = \frac{1}{Z} \int_m^M \frac{\theta}{\theta^n} \, d\theta \ \xrightarrow{M \to \infty} \ m \, \frac{n-1}{n-2}, \qquad p(x \mid T) = \int_{\max(x,m)}^{M} \frac{1}{\theta} \, p(\theta \mid T) \, d\theta$$

ML, MAP, MMSE and Bayes for the tanks



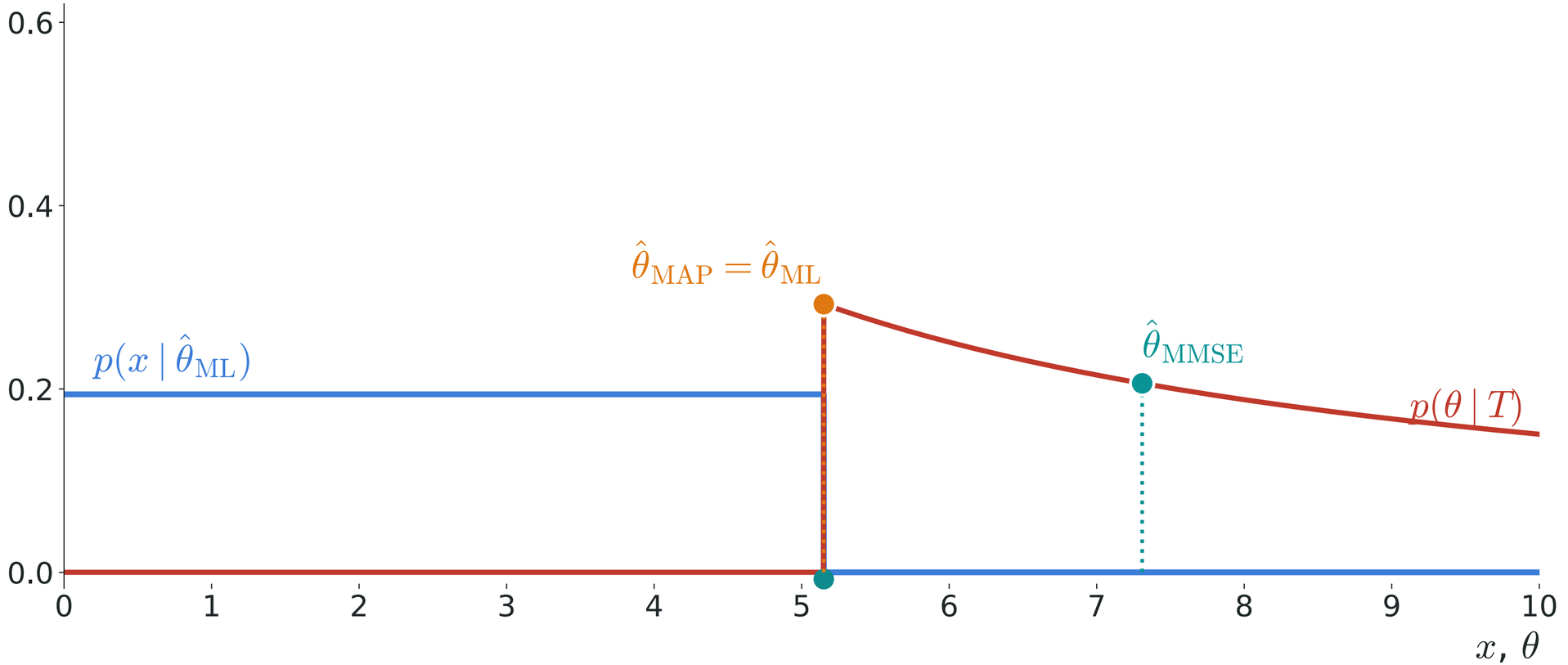
One tank, $x_1 = 5.15$: the ML plug-in density ends at the observation.

ML, MAP, MMSE and Bayes for the tanks



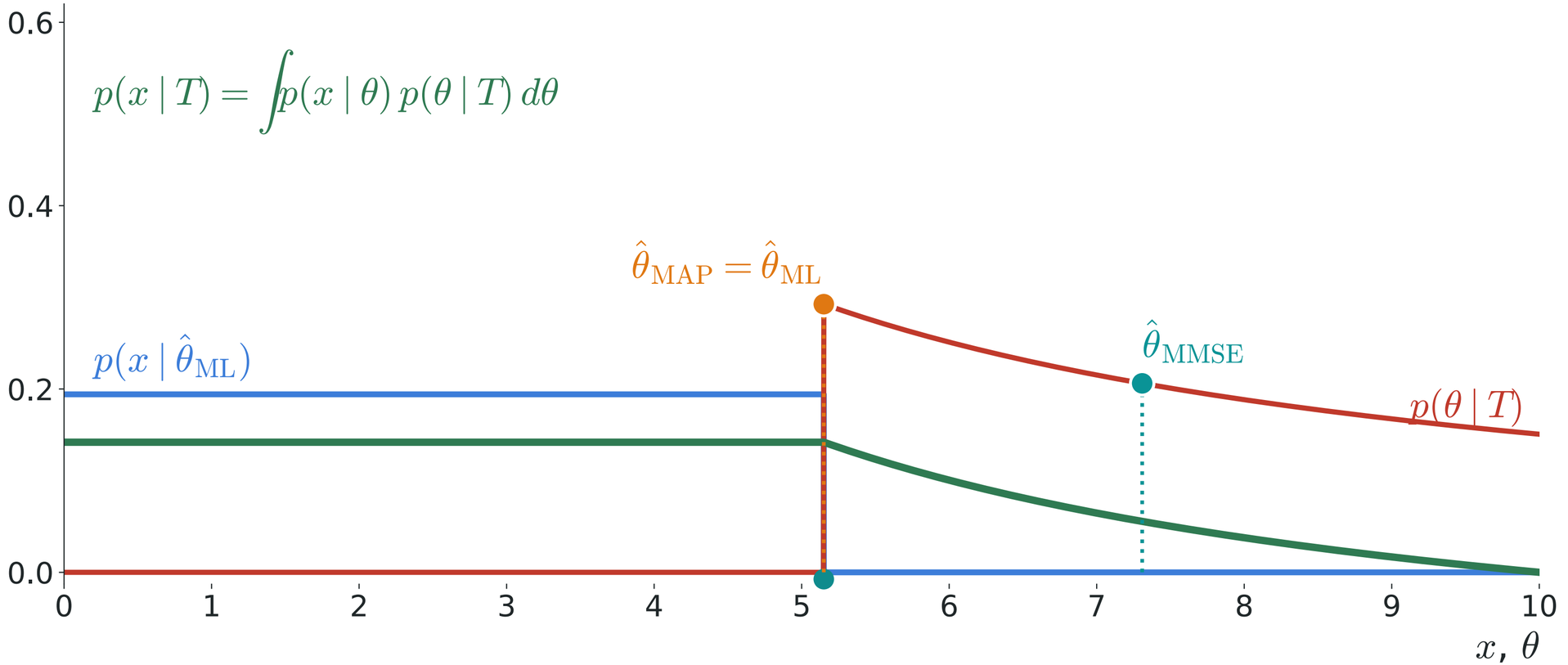
Posterior over θ (prior uniform on $[0, 10]$): its maximum is again m.

ML, MAP, MMSE and Bayes for the tanks



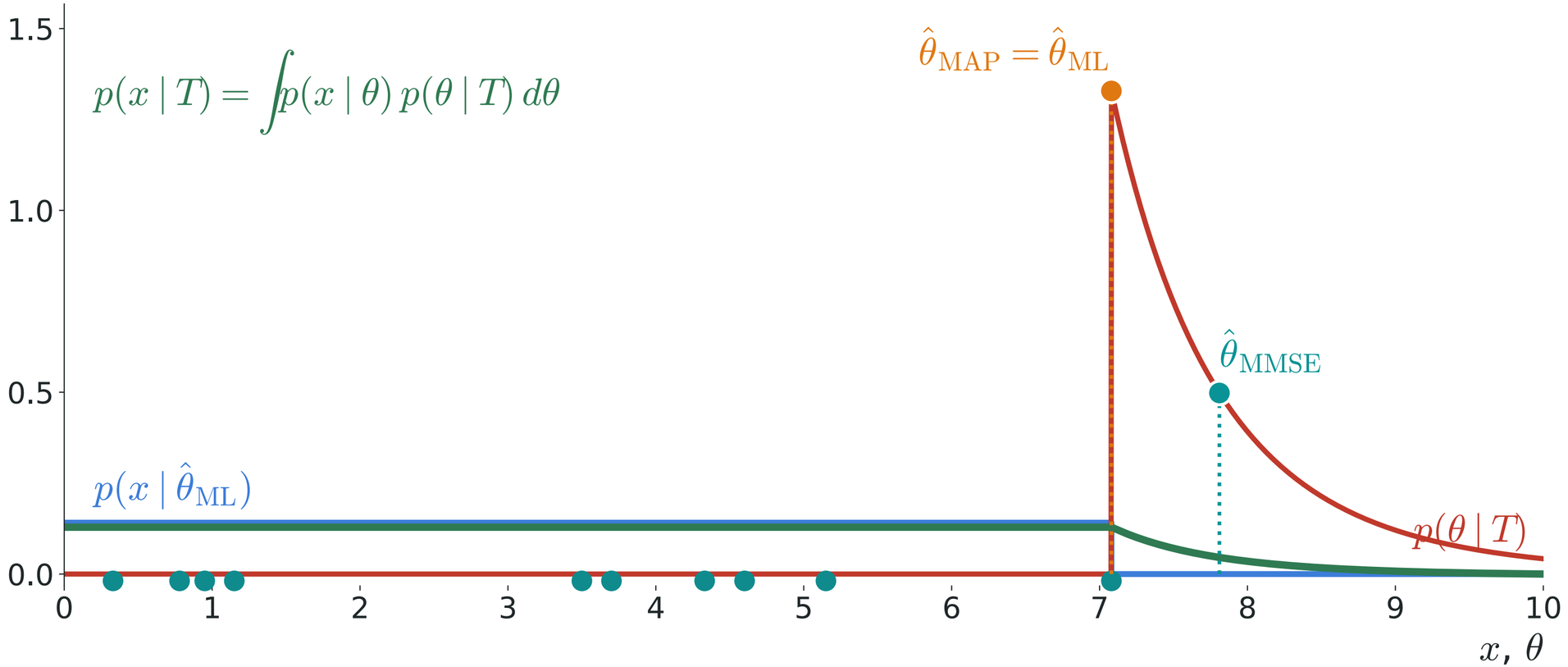
Its mean $\hat{\theta}_{\mathrm{MMSE}} = 7.3$ lies to the right of m.

ML, MAP, MMSE and Bayes for the tanks



Bayesian prediction $p(x \mid T)$: larger numbers are still possible.

ML, MAP, MMSE and Bayes for the tanks



Ten tanks: the posterior is narrow, all estimates are close.

German tanks: try it

n tanks 1

true θ 7.3

prior M 10

show
- [x] ML plug-in
- [ ] posterior
- [ ] MMSE
- [ ] predictive

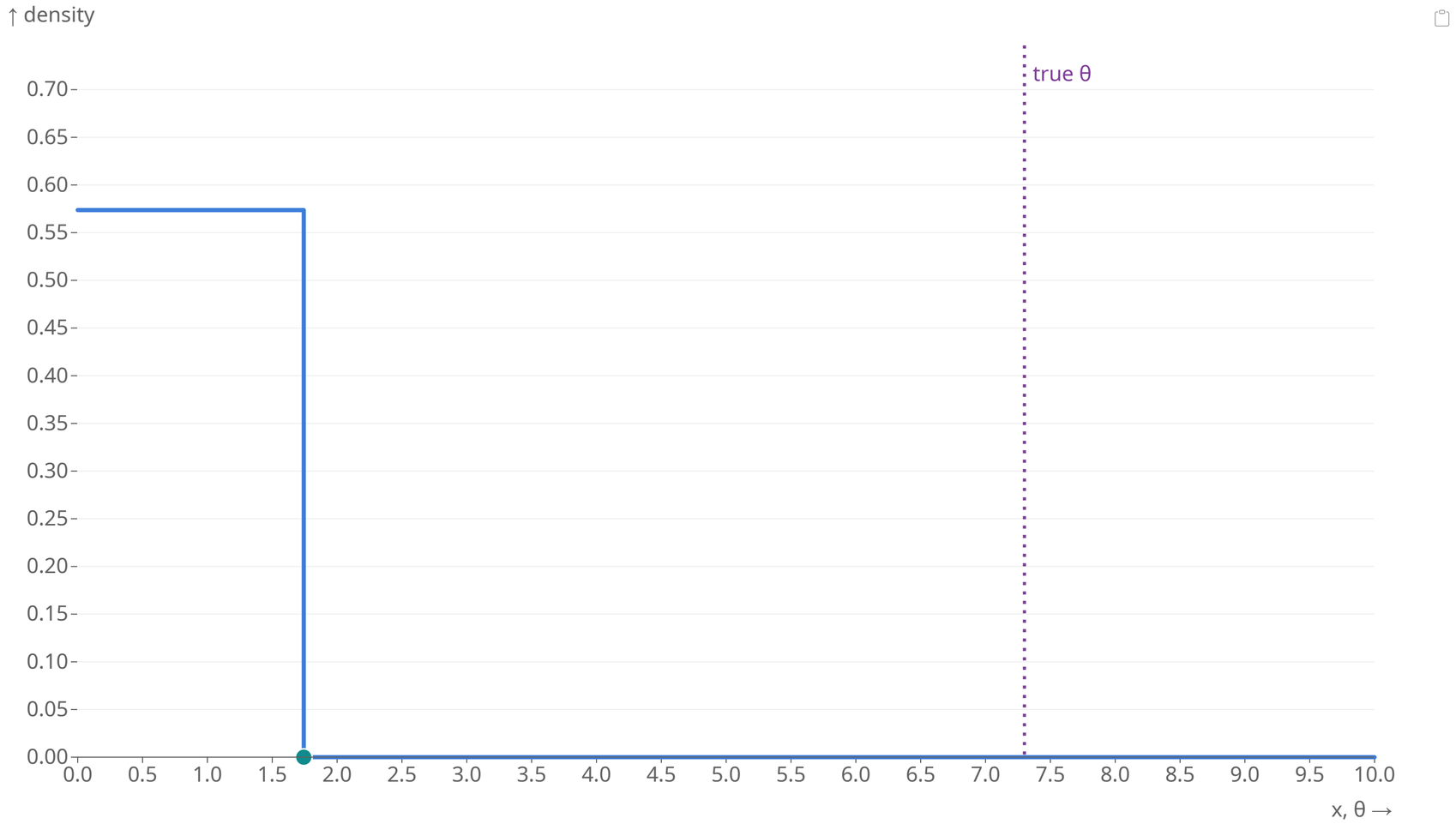
new tanks

$\theta_{ML} = \theta_{MAP} = 1.74$

$\theta_{MMSE} = 4.73$

unbiased m(n+1)/n = 3.49

P(next x > m) = 0.528



The real (discrete) problem

Serial numbers $1, 2, \ldots, N$, we see k of them, the largest is m:

estimate	formula	$k = 4, m = 60$
ML	m	60
unbiased	$m + \frac{m}{k} - 1$	74
Bayes, posterior mean	$\frac{(m-1)(k-1)}{k-2}$	89

ML is **biased**: it always underestimates. Adding the average gap $\frac{m}{k}$ fixes it.

Why is **ML** the most common?

Why ML?

Under general conditions, as $N \to \infty$

- **consistent**: converges to the true θ
- asymptotically **unbiased** and **efficient** (the smallest variance)
- asymptotically **normal**

In practice

- no prior to choose
- a lot of data: MAP and Bayes give the **same** result
- Bayes needs integrals: closed form only for conjugate priors

But with few data

- overconfident: tanks (and socks, unseen categories later)
- biased: $\hat{\sigma}^2_{\mathrm{ML}} = \frac{1}{N}\sum(x_i - \hat{\mu})^2$, $\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}] = \frac{N-1}{N}\sigma^2$

A bit of history

- **1769** Lagrange: ML for the multinomial gives relative frequencies
- **1778** D. Bernoulli: multiply the densities of the observations, maximize
- **1809** Gauss: posterior mode with a flat prior, gives **least squares**
- **1912, 1922** Fisher: the method gets its name and its theory

> "Who would rise to argue in favor of a method of minimum likelihood, or even mediocre likelihood?"

S. M. Stigler: *The Epic Story of Maximum Likelihood*, Statistical Science, 2007.

Example: socks

Socks

A drawer with **red** and **green** socks. We pull out a sock, note its colour and **put it back**. After N draws, R socks were red.

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{\text{red}, \text{green}\}$$

$$p(\text{red} \mid \pi) = \pi, \qquad p(\text{green} \mid \pi) = 1 - \pi$$

π: the ratio of red socks in the drawer

Worked out on the board in the lab.

?

1. What is the likelihood $L(\pi)$?
2. Find $\hat{\pi}_{\text{ML}}$.
3. Find $\hat{\pi}_{\text{MAP}}$ with a prior $\beta(\pi \mid a, b)$.
4. Does it matter if T is an **ordered** sequence?

More problems like this: exercise book on CourseWare.

Socks

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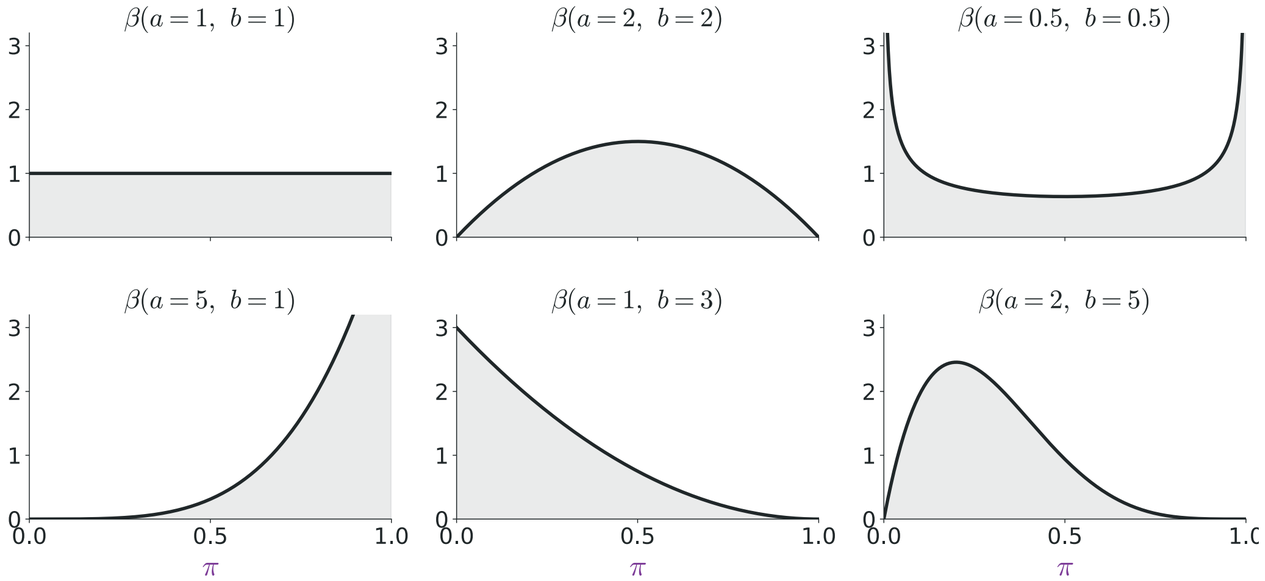
$$\hat{\pi}_{\text{ML}} = \frac{R}{N}, \qquad \hat{\pi}_{\text{MAP}} = \frac{R + a - 1}{N + a + b - 2}$$

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The Beta prior: virtual observations

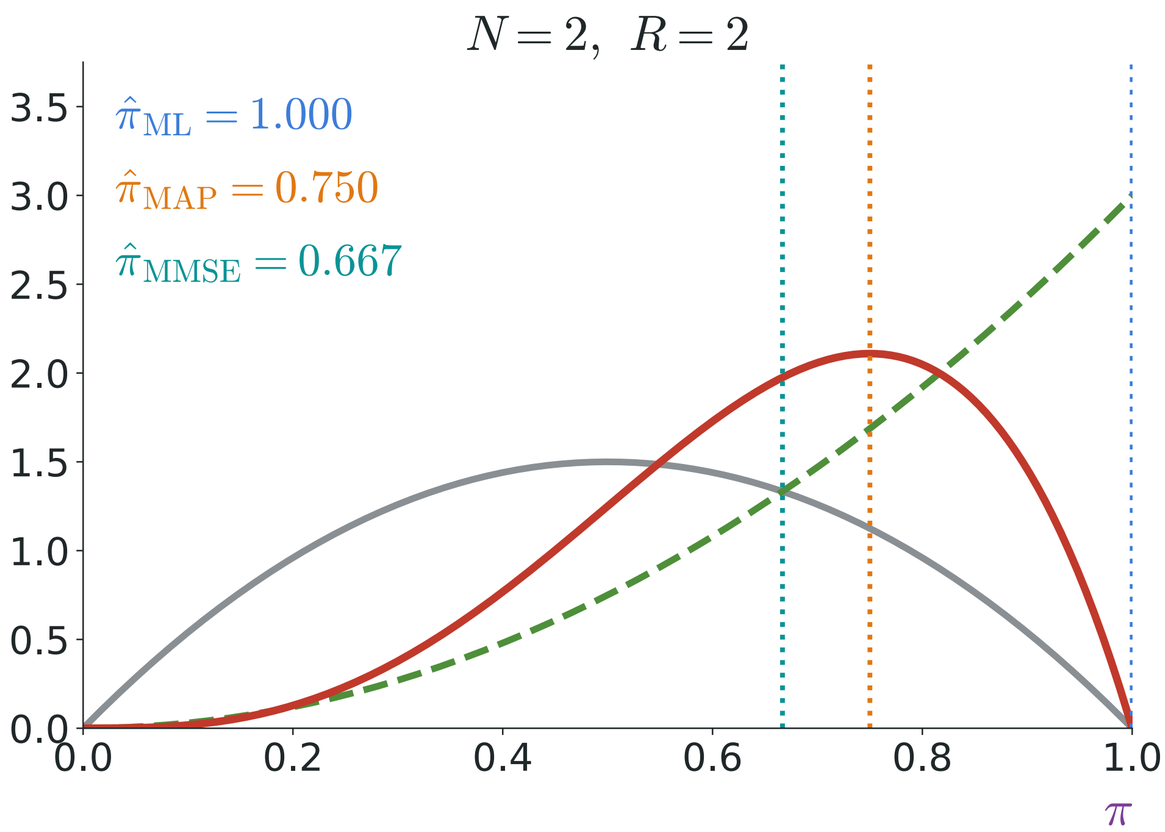


$$\hat{\pi}_{\text{MAP}} = \frac{R + a - 1}{N + a + b - 2}$$

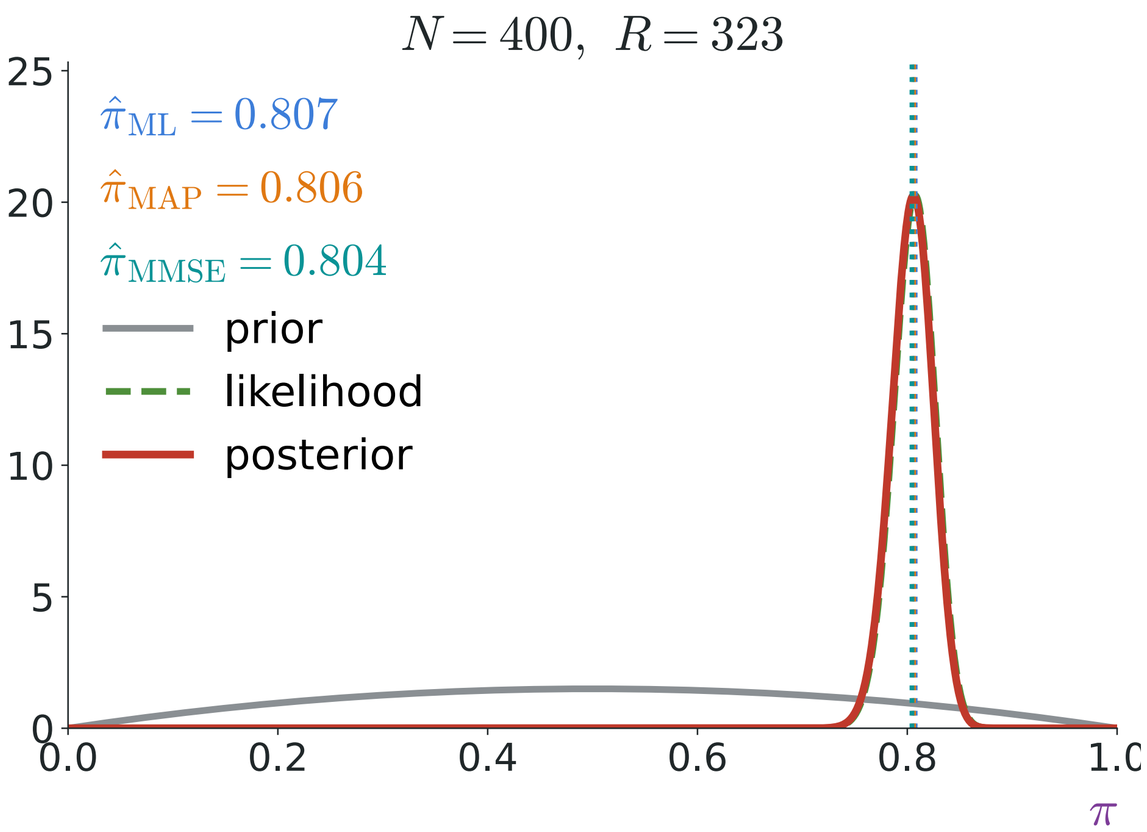
> ! As if we had seen $a - 1$ extra **red** and $b - 1$ extra **green** socks.

a, b are **hyperparameters**: parameters of the prior.

Socks in numbers



Two red socks: ML says **all** socks are red.



400 socks: the prior **does not matter** any more.

Socks: try it

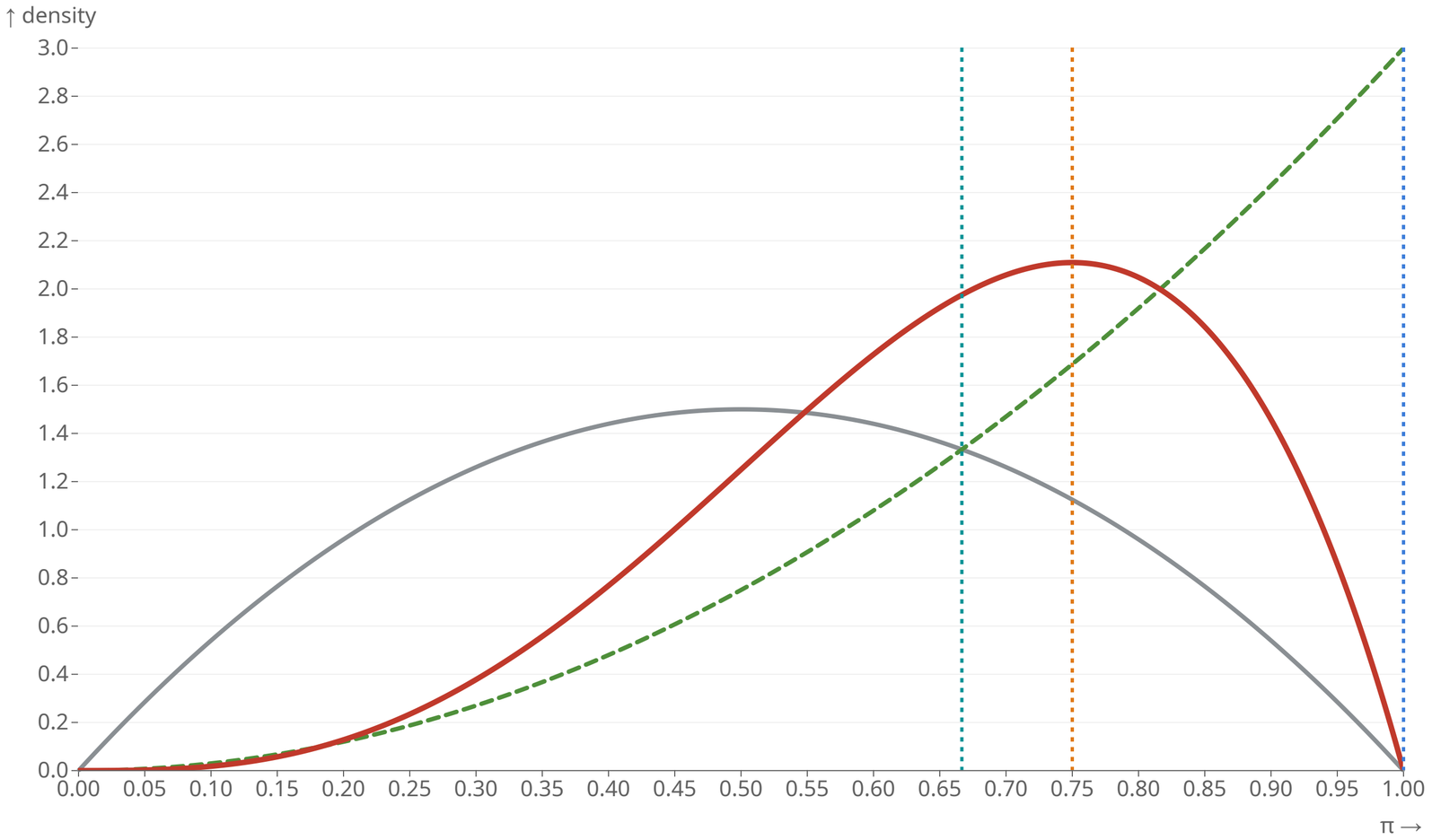
N socks	2
R red	2
prior a	2
prior b	2

$\pi_{ML} = 1.000$

$\pi_{MAP} = 0.750$

$\pi_{MMSE} = 0.667$

prior, likelihood, posterior



Categorical distribution

A die with K sides

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{1, \ldots, K\}$$

$$p(x = k \mid p_1, \ldots, p_K) = p_k$$

$$\sum_{k=1}^{K} p_k = 1, \qquad N_k = \text{count of } k$$

Worked out on the board in the lab.

? 1. Write the likelihood $L(p_1, \ldots, p_K)$.
2. Find $\hat{p}_k^{\mathrm{ML}}$. Do not forget the **constraint**!

Hint: use a **Lagrange multiplier** for the constraint $\sum_k p_k = 1$.

More problems like this: exercise book on CourseWare.

A biased die, an election, words in a text.

A die with K sides

$$T = \{x_1, \ldots, x_N\}, \quad x_i \in \{1, \ldots, K\}$$

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Worked out on the board in the lab.

$$\hat{p}_k^{\mathrm{ML}} = \frac{N_k}{N} \text{ (the relative frequency)}$$

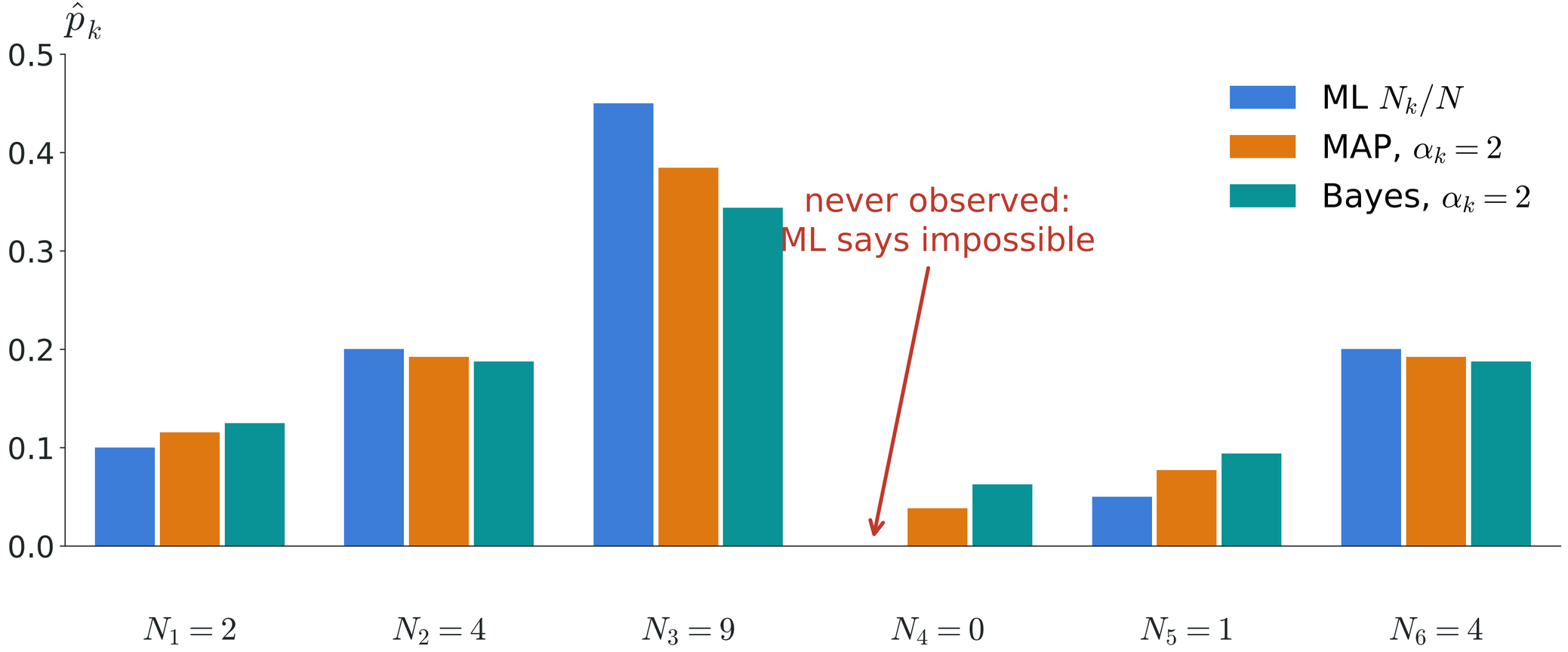
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Hint: use a **Lagrange multiplier** for the constraint $\sum_k p_k = 1$.

More problems like this: exercise book on CourseWare.

A biased die, an election, words in a text.

A category never observed



ML gives an unseen category probability **0**. The prior gives it a small chance.

Practice

Exponential distribution

Waiting times $T = \{t_1, \ldots, t_N\}$, exponential distribution:

$$p(t \mid \lambda) = \frac{1}{\lambda} e^{-t/\lambda}, \qquad t \geq 0$$

λ: the mean waiting time

Try it yourself, then check your result:

? 1. Find $\hat{\lambda}_{\mathrm{ML}}$.
2. Substitute $\theta = 1/\lambda$ and find $\hat{\theta}_{\mathrm{ML}}$. What is the relation?

More problems like this: exercise book on CourseWare.

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Try it yourself, then check your result:

$$\hat{\lambda}_{\mathrm{ML}} = \frac{1}{N} \sum_i t_i, \qquad \hat{\theta}_{\mathrm{ML}} = \frac{1}{\hat{\lambda}_{\mathrm{ML}}}$$

1. Find $\hat{\lambda}_{\mathrm{ML}}$.
2. Substitute $\theta = 1/\lambda$ and find $\hat{\theta}_{\mathrm{ML}}$. What is the relation?

More problems like this: exercise book on CourseWare.

The assignment

Assignment: MLE, MAP, Bayes

Part 1: normal

- ML of μ, σ^2
- MAP: **normal-inverse-gamma** prior
- Bayes: posterior and **predictive** density

Part 2: categorical

- ML: relative frequencies
- MAP: **Dirichlet** prior
- Bayes: predictive $\frac{N_k+\alpha_k}{N+\sum_j \alpha_j}$

Part 3: classifier

Letters **A and C**, training sets of 20, 200 and 2000 samples.

Compare ML, MAP and Bayes on the test set.

Take-away for the lab

- ☑ Known form $p(x \mid \theta)$: estimate the **parameters** from i.i.d. samples
- ☑ **ML**: $\arg\max_\theta p(T \mid \theta)$, simple, the best with a lot of data
- ☑ **MAP**: $\arg\max_\theta p(T \mid \theta)\, p(\theta)$, the prior as **virtual observations**
- ☑ **Bayes**: the whole posterior, prediction **averages** all models
- ☑ **MMSE**: the mean of the posterior
- ☑ With few data the choice **matters**: socks, unseen categories, German tanks


"That's all Folks!"